ASKNDER NADA

IDSS

2025

Uninformed Search Strategy

1. Which search strategy is also called as blind search?

a) Uninformed search b) Informed search c) Simple reflex search d) All of the mentioned

Answer: A

Explanation: In blind search, We can search the states without having any additional information. So uninformed search method is blind search.

2. How many types are available in uninformed search method?

a) 3 b) 4 c) 5 d) 6

Answer: c

Explanation: The five types of uninformed search method are Breadth-first, Uniform-cost, Depth-first, Depth-limited and Bidirectional search.

3. Which search is implemented with an empty first-in-first-out queue?

a) Depth-first search b) Breadth-first search c) Bidirectional search d) None of the mentioned

Answer: b

Explanation: Because of FIFO queue, it will assure that the nodes that are visited first will be expanded first.

4. When is breadth-first search is optimal?

a) When there is less number of nodes b) When all step costs are equal

c) When all step costs are unequal d) None of the mentioned

answer: b

Explanation: Because it always expands the shallowest unexpanded node.

5. How many parts does a problem consists of?

a) 1 b) 2 c) 3 d) 4

Answer: d

Explanation: The four parts of the problem are initial state, set of actions, goal test and path cost.

6. Which algorithm is used to solve any kind of problem?

a) Breadth-first algorithm b) Tree algorithm c) Bidirectional search algorithm d) None of the mentioned

Answer: b

Explanation: Tree algorithm is used because specific variants of the algorithm embed different strategies.

7. Which search algorithm imposes a fixed depth limit on nodes?

a) Depth-limited search b) Depth-first search

c) Iterative deepening search d) Bidirectional search

Answer: a

Explanation: None.

8. Which search implements stack operation for searching the states?

a) Depth-limited search b) Depth-first search c) Breadth-first search d) None of the mentioned

Answer: b

Explanation: It implements stack operation because it always expands the deepest node in the current tree.

1. What is the general term of Blind searching?
a) Informed Search b) Uninformed Search c) Informed & Unformed Search d) Heuristic Search

Answer: b
Explanation: In case of uninformed search no additional information except the problem definition is given.

2. Strategies that know whether one non-goal state is "more promising" than another are called ___________
a) Informed & Unformed Search b) Unformed Search
c) Heuristic & Unformed Search d) Informed & Heuristic Search

Answer: d
Explanation: Strategies that know whether one non-goal state is "more promising" than another are called informed search or heuristic search strategies.

3. Which of the following is/are Uninformed Search technique/techniques?
a) Breadth First Search (BFS) b) Depth First Search (DFS)
c) Bidirectional Search d) All of the mentioned

Answer: d
Explanation: Several uninformed search techniques includes BFS, DFS, Uniform-cost, Depth-limited, Bidirectional search etc.

4. Which data structure conveniently used to implement BFS?
a) Stacks b) Queues c) Priority Queues d) All of the mentioned

Answer: b
Explanation: Queue is the most convenient data structure, but memory used to store nodes can be reduced by using circular queues.

5. Which data structure conveniently used to implement DFS?
a) Stacks b) Queues c) Priority Queues d) All of the mentioned

Answer: a
Explanation: DFS requires node to be expanded the one most recent visited, hence stack is convenient to implement.

6. Breadth-first search is not optimal when all step costs are equal, because it always expands the shallowest unexpanded node.
a) True b) False

Answer: b
Explanation: Breadth-first search is optimal when all step costs are equal, because it always expands the shallowest unexpanded node. If the solution exists in shallowest node no irrelevant nodes are expanded.

7. uniform-cost search expands the node n with the __________
a) Lowest path cost b) Heuristic cost c) Highest path cost d) Average path cost

Answer: a
Explanation: Uniform-cost search expands the node n with the lowest path cost. Note that if all step costs are equal, this is identical to breadth-first search.

8. Depth-first search always expands the ______ node in the current fringe of the search tree.
a) Shallowest b) Child node c) Deepest d) Minimum cost

Answer: c
Explanation: Depth-first search always expands the deepest/leaf node in the current fringe of the search tree.

9. Breadth-first search always expands the ______ node in the current fringe of the search tree.
a) Shallowest b) Child node c) Deepest d) Minimum cost

Answer: a
Explanation: Breadth-first search always expands the shallowest node in the current fringe of the search tree. Traversal is performed level wise.

11. Optimality of BFS is ___________
a) When there is less number of nodes b) When all step costs are equal
c) When all step costs are unequal d) None of the mentioned

Answer: b
Explanation: It always expands the shallowest unexpanded node.

DFS BFS
12. LIFO is ______ where as FIFO is ________
a) Stack, Queue b) Queue, Stack c) Priority Queue, Stack d) Stack. Priority Queue

Answer: a
Explanation: LIFO is last in first out – Stack. FIFO is first in first out – Queue..

13. For general graph, how one can get rid of repeated states?
a) By maintaining a list of visited vertices b) By maintaining a list of traversed edges
c) By maintaining a list of non-visited vertices d) By maintaining a list of non-traversed edges

Answer: a
Explanation: Other techniques are costly.

14. DFS is ______ efficient and BFS is __________ efficient.
a) Space, Time b) Time, Space c) Time, Time d) Space, Space

Answer: a

Informed Search Strategy

1. What is the other name of informed search strategy?
a) Simple search b) Heuristic search c) Online search d) None of the mentioned

Answer: b
Explanation: A key point of informed search strategy is heuristic function, So it is called as heuristic function.

2. How many types of informed search method are in artificial intelligence?
a) 1 b) 2 c) 3 d) 4

Answer: d
1 2 3
Explanation: The four types of informed search method are best-first search, Greedy best-first search, A* search and memory bounded heuristic search.
4

3. Which search uses the problem specific knowledge beyond the definition of the problem?
a) Informed search b) Depth-first search c) Breadth-first search d) Uninformed search

Answer: a
Explanation: Informed search can solve the problem beyond the function definition, So does it can find the solution more efficiently.

4. What is the heuristic function of greedy best-first search?
a) f(n) != h(n) b) f(n) < h(n) c) f(n) = h(n) d) f(n) > h(n)

Answer: c.

5. Which search is complete and optimal when h(n) is consistent?
a) Best-first search b) Depth-first search c) Both Best-first & Depth-first search d) A* search

Answer: d
Explanation: None.

6. Which is used to improve the performance of heuristic search?
a) Quality of nodes b) Quality of heuristic function c) Simple form of nodes d) None of the mentioned

Answer: b
Explanation: Good heuristic can be constructed by relaxing the problem, So the performance of heuristic search can be improved.

7. Which search method will expand the node that is closest to the goal?
a) Best-first search b) Greedy best-first search c) A* search d) None of the mentioned

Answer: b
Explanation: Because of using greedy best-first search, It will quickly lead to the solution of the problem.

1. A heuristic is a way of trying ___________
a) To discover something or an idea embedded in a program
b) To search and measure how far a node in a search tree seems to be from a goal
c) To compare two nodes in a search tree to see if one is better than another
d) All of the mentioned

Answer: d
Explanation: In a heuristic approach, we discover certain idea and use heuristic functions to search for a goal and predicates to compare nodes.

2. The search strategy the uses a problem specific knowledge is known as ___________
a) Informed Search b) Best First Search c) Heuristic Search d) All of the mentioned

Answer: d
Explanation: The problem specific knowledge is also known as Heuristics and Best-First search uses some heuristic to choose the best node for expansion.

3. Uninformed search strategies are better than informed search strategies.
a) True b) False

Answer: b
Explanation: Informed search strategies uses some problem specific knowledge, hence more efficient to finding goals.

4. Heuristic function h(n) is ________

a) Lowest path cost b) Cheapest path from root to goal node

c) Estimated cost of cheapest path from root to goal node d) Average path cost

Answer: c

Explanation: Heuristic is an estimated cost.

5. Greedy search strategy chooses the node for expansion in ___________

a) Shallowest b) Deepest c) The one closest to the goal node d) Minimum heuristic cost

Answer: c

Explanation: Sometimes minimum heuristics can be used, sometimes maximum heuristics function can be used. It depends upon the application on which the algorithm is applied.

6. What is the evaluation function in greedy approach?

a) Heuristic function b) Path cost from start node to current node

c) Path cost from start node to current node + Heuristic cost

d) Average of Path cost from start node to current node and Heuristic cost

Answer: a

Explanation: Greedy best-first search3 tries to expand the node that is closest to the goal, on the grounds that this is likely to lead to a solution quickly. Thus, it evaluates nodes by using just the heuristic function: f (n) = h(n).

7. What is the evaluation function in A* approach?

a) Heuristic function b) Path cost from start node to current node

c) Path cost from start node to current node + Heuristic cost

d) Average of Path cost from start node to current node and Heuristic cost

Answer: c

Explanation: The most widely-known form of best-first search is called A* search. It evaluates nodes by combining g(n), the cost to reach the node, and h(n.), the cost to get from the node to the goal: f(n) = g(n) + h(n). Since g(n) gives the path cost from the start node to node n, and h(n) is the estimated cost of the cheapest path from n to the goal.

8. A* is optimal if h(n) is an admissible heuristic-that is, provided that h(n) never underestimates the cost to reach the goal.

a) True b) False

Answer: a

Explanation: A* is optimal if h(n) is an admissible heuristic-that is, provided that h(n) never overestimates the cost to reach the goal. Refer both the example from the book for better understanding of the algorithms.

Adversarial Search Test

1. Zero sum game has to be a ______ game.

a) Two player B) Multiplayer C) Three player D) Single player

Answer:B

Explanation: Zero sum games could be multiplayer games as long as the condition for zero sum game is satisfied.

2. Mathematical game theory, a branch of economics, views any multi-agent environment as a game provided that the impact of each agent on the others is "significant," regardless of whether the agents are cooperative or competitive.

A) False B) True

Answer : B

3. General algorithm applied on game tree for making decision of win/lose is ___________

A) DFS/BFS Search Algorithms B) Heuristic Search Algorithms
C) MIN/MAX Algorithms D) Greedy Search Algorithms

Answer: C

4. A game can be formally defined as a kind of search problem with the following components.

A) Successor Function B) Initial State C) All of the mentioned D)Terminal Test

Answer: C

5. The minimax algorithm computes the minimax decision from the current state. It uses a simple recursive computation of the minimax values of each successor state, directly implementing the defining equations. The recursion proceeds all the way down to the leaves of the tree, and then the minimax values are backed up through the tree as the recursion unwinds.

A) True B) False

Answer: A

6. Adversarial search problems uses ___________

A) Only Competitive and Cooperative Environment B) Neither Competitive nor Cooperative Environment

C) Competitive Environment D) Cooperative Environment

Answer: C

7. Zero sum games are the one in which there are two agents whose actions must alternate and in which the utility values at the end of the game are always the same.

A) False B) True

Answer: A

Explanation: Utility values are always same and opposite.

8. The initial state and the legal moves for each side define the _________ for the game.

A) Game Tree B) State Space Search C) Search Tree D) Forest

Answer: A

Explanation: An example of game tree for Tic-Tac-Toe game.

9. General games involves ___________

A) Multi-agent B) Neither Single-agent nor Multi-agent

C) Only Single-agent and Multi-agent D) Single-agent

Answer: C

10. What is the complexity of minimax algorithm?

A) Time – bm and space – bm B) Same as BFS

C) Space – bm and time – bm D) Same as of DFS

Alpha Beta Pruning

1. Which search is equal to minimax search but eliminates the branches that can't influence the final decision?
a) Depth-first search b) Breadth-first search c) Alpha-beta pruning d) None of the mentioned

Answer: c
Explanation: The alpha-beta search computes the same optimal moves as minimax, but eliminates the branches that can't influence the final decision.

2. Which values are independant in minimax search algorithm?
a) Pruned leaves x and y b) Every states are dependant
c) Root is independent d) None of the mentioned

Answer: a
Explanation: The minimax decision are independant of the values of the pruned values x and y because of the root values.

3. To which depth does the alpha-beta pruning can be applied?
a) 10 states b) 8 States c) 6 States d) Any depth

Answer: d
Explanation: Alpha–beta pruning can be applied to trees of any depth and it is possible to prune entire subtree rather than leaves.

4. Which search is similar to minimax search?
a) Hill-climbing search b) Depth-first search c) Breadth-first search d) All of the mentioned

Answer: b
Explanation: The minimax search is depth-first search, So at one time we just have to consider the nodes along a single path in the tree.

5. Which value is assigned to alpha and beta in the alpha-beta pruning?
a) Alpha = max b) Beta = min c) Beta = max d) Both Alpha = max & Beta = min

Answer: d
Explanation: Alpha and beta are the values of the best choice we have found so far at any choice point along the path for MAX and MIN.

6. Where does the values of alpha-beta search get updated?
a) Along the path of search b) Initial state itself
c) At the end d) None of the mentioned

Answer: a
Explanation: Alpha-beta search updates the value of alpha and beta as it gets along and prunes the remaining branches at node.

7. How the effectiveness of the alpha-beta pruning gets increased?
a) Depends on the nodes
b) Depends on the order in which they are executed
c) All of the mentioned
d) None of the mentioned

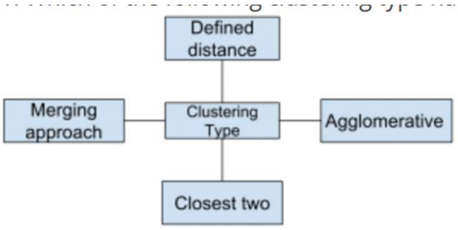
Answer: a
Explanation: None.

8. Which function is used to calculate the feasibility of whole game tree?
a) Evaluation function
b) Transposition
c) Alpha-beta pruning
d) All of the mentioned

Answer: a
Explanation: Because we need to cut the search off at some point and apply an evaluation function that gives an estimate of the utility of the state.

Clustering

1. Which of the following clustering type has characteristic shown in the below figure



a) Partitional
b) Hierarchical
c) Naive bayes
d) None of the mentioned

Answer: b
Explanation: Hierarchical clustering groups data over a variety of scales by creating a cluster tree or dendrogram.

2. Point out the correct statement.
a) The choice of an appropriate metric will influence the shape of the clusters
b) Hierarchical clustering is also called HCA
c) In general, the merges and splits are determined in a greedy manner
d) All of the mentioned

Answer: d
Explanation: Some elements may be close to one another according to one distance and farther away according to another.

3. Which of the following is finally produced by Hierarchical Clustering?
a) final estimate of cluster centroids
b) tree showing how close things are to each other
c) assignment of each point to clusters
d) all of the mentioned

Answer: b
Explanation: Hierarchical clustering is an agglomerative approach.

4. Which of the following is required by K-means clustering?
a) defined distance metric
b) number of clusters
c) initial guess as to cluster centroids
d) all of the mentioned

Answer: d
Explanation: K-means clustering follows partitioning approach.

5. Point out the wrong statement.
a) k-means clustering is a method of vector quantization
b) k-means clustering aims to partition n observations into k clusters
c) k-nearest neighbor is same as k-means d) none of the mentioned

Answer: c
Explanation: k-nearest neighbor has nothing to do with k-means.

6. Which of the following combination is incorrect?
a) Continuous – euclidean distance b) Continuous – correlation similarity
c) Binary – manhattan distance d) None of the mentioned

Answer: d
Explanation: You should choose a distance/similarity that makes sense for your problem.

7. Hierarchical clustering should be primarily used for exploration.
a) True b) False

Answer: a
Explanation: Hierarchical clustering is deterministic.

8. Which of the following function is used for k-means clustering?
a) k-means b) k-mean c) heatmap d) none of the mentioned

Answer: a
Explanation: K-means requires a number of clusters.

9. Which of the following clustering requires merging approach?
a) Partitional b) Hierarchical c) Naive Bayes d) None of the mentioned

Answer: b
Explanation: Hierarchical clustering requires a defined distance as well.

10. K-means is not deterministic and it also consists of number of iterations.
a) True b) False

Answer: a
Explanation: K-means clustering produces the final estimate of cluster centroids.

1. : What is the primary objective of k-Means clustering in data mining?
(A) Classification B) Regression C) Clustering (D) Association rule mining

Answer: c

2. : How does the k-Means algorithm initialize cluster centroids?
(A) Randomly (B) Using the mean of all data points
(C) Based on the median data point (D) By choosing the farthest data points

Answer: a

3. : What is the role of the 'k' parameter in the k-Means algorithm?
(A) Number of clusters to be formed (B) Distance metric used for clustering
(C) Learning rate for centroid updates (D) Number of iterations for convergence

Answer: a

4. : How does the k-Means algorithm update cluster centroids during each iteration?
(A) By calculating the mean of all data points in each cluster B) By choosing the data point closest to the centroid
(C) By merging clusters with similar centroids (D) By selecting the most central data point

Answer: a

5. : What is a major limitation of the k-Means algorithm?
(A) It cannot handle large datasets (B) It requires a large number of clusters
(C) It is sensitive to initial centroid positions (D) It cannot handle categorical data

Answer: c

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6. : How does the k-Means algorithm determine convergence?
(A) When the centroids stop moving significantly between iterations
(B) When all data points are assigned to a cluster
(C) After a fixed number of iterations (D) When the number of clusters equals ‘k’

Answer: a

7. : Which distance metric is commonly used in the k-Means algorithm?
(A) Manhattan distance (B) Hamming distance (C) Euclidean distance (D) Cosine similarity

Answer: c

8. : Which of the following methods can help improve the performance of the k-Means algorithm?
(A) Using a larger value of ‘k’ (B) Reducing the number of iterations
(C) Scaling the data to have equal variance (D) Initializing centroids close to the mean of the data

Answer: c

9. : What is the main advantage of the k-Means algorithm?
(A) It is robust to outliers (B) It guarantees convergence to the global optimum
(C) It can handle non-linear data (D) It is computationally efficient

Answer: D

What happens if the number of specified clusters in K-means clustering is too small?

A) The algorithm will fail to converge
B) The resulting clusters will be too broad and may not capture the underlying structure of the data
C) The resulting clusters will be too specific and may overfit the data
D)The algorithm will automatically adjust the number of clusters

Answer: B.
Explanation: If the number of specified clusters in K-means clustering is too small, the resulting clusters will be too broad and may not capture the underlying structure of the data, leading to suboptimal clustering solutions.

In K-means clustering, which of the following techniques can be used to address the sensitivity to the initial placement of cluster centroids?

A)Random initialization B) K-means++
C) Running the algorithm multiple times with different initializations and selecting the best solution
D) Both B and C

Correct Answer: D.

Explanation: To address the sensitivity to the initial placement of cluster centroids in K-means clustering, both K-means++ initialization and running the algorithm multiple times with different initializations can be used. K-means++ improves the initial placement of centroids, while running the algorithm multiple times increases the likelihood of finding a better clustering solution by selecting the solution with the lowest within-cluster sum of squares.

How can K-means clustering be extended to handle mixed-type data (both continuous and categorical)?

A) By using the Gower distance
B) By one-hot encoding the categorical variables
C) By standardizing the continuous variables
D) All of the above

Correct Answer: D.
Explanation:K-means clustering can be extended to handle mixed-type data by using the Gower distance, one-hot encoding categorical variables, and standardizing continuous variables. These preprocessing steps can help account for the differences between continuous and categorical data.

Which of the following clustering algorithms can be used as an alternative to K-means clustering for handling categorical data?

A) DBSCAN
B) Hierarchical clustering
C) K-modes
D) Spectral clustering

Correct Answer: C.
Explanation:K-modes is a clustering algorithm specifically designed to handle categorical data. It is an alternative to K-means clustering and replaces the mean-based centroid calculation with a mode-based calculation.

Which of the following is a limitation of K-means clustering in handling imbalanced datasets?

A) K-means assumes that all clusters have similar sizes
B) K-means is sensitive to outliers
C) K-means assumes that all clusters have a spherical shape
D) K-means is sensitive to the initial placement of cluster centroids

Correct Answer: A.
Explanation:K-means clustering assumes that all clusters have similar sizes, which can be a limitation when handling imbalanced datasets, as the algorithm may not perform well on clusters with significantly different sizes.

How can K-means clustering be used for outlier detection?

A) By identifying data points that are far from their cluster centroids
B) By identifying clusters with few data points
C) By identifying data points that have a high silhouette score
D) By identifying data points with a high between-cluster variance

Correct Answer:A. By identifying data points that are far from their cluster centroids

Explanation:K-means clustering can be used for outlier detection by identifying data points that are far from their cluster centroids. These data points may be considered outliers, as they do not fit well within their assigned cluster.

In K-means clustering, which of the following factors can impact the quality of the clustering solution?

A) The number of clusters
B) The distance metric used
C) The initialization method
D) All of the above

Correct Answer: D.
Explanation: All of the listed factors can impact the quality of the clustering solution in K-means clustering: the number of clusters, the distance metric used, and the initialization method.

In K-means clustering, how are initial centroids typically selected?

A) Randomly from the data points
B) By using the K-means++ initialization method
C) By employing a separate clustering algorithm
D) Both A and B

Correct Answer:D. Both A and B
Explanation: In K-means clustering, initial centroids are typically selected either randomly from the data points or by using the K-means++ initialization method, which reduces the sensitivity to the initial placement of cluster centroids.

What is the difference between K-means clustering and hierarchical clustering?

A) K-means is a partitional clustering method, while hierarchical clustering is a tree-based method
B) K-means is sensitive to outliers, while hierarchical clustering is robust to outliers
C) K-means requires the number of clusters to be specified, while hierarchical clustering does not
D) All of the above

Correct Answer: D.

What happens if the number of specified clusters in K-means clustering is too large?

A) The algorithm will fail to converge B) The resulting clusters will be too specific and may overfit the data
C)The resulting clusters will be too broad and may not capture the underlying structure of the data
D) The algorithm will automatically adjust the number of clusters

Correct Answer: B
Explanation: If the number of specified clusters in K-means clustering is too large, the resulting clusters will be too specific and may overfit the data, leading to suboptimal clustering solutions.

Which of the following is a common application of K-means clustering?

A) Image segmentation B) Text classification C) Anomaly detection D) Time series forecasting

Correct Answer: A.
Explanation: Image segmentation is a common application of K-means clustering, as it involves partitioning an image into regions based on pixel intensities or colors.

What type of data does K-means clustering work best with?

A) Continuous data B) Categorical data C) Binary data D) Text data

Correct Answer:A. Continuous data

Explanation:K-means clustering works best with continuous data, as it relies on Euclidean distance to measure similarity between data points.

Which of the following is a disadvantage of the K-means clustering algorithm?

A) It assumes clusters have a spherical shape
B) It cannot handle categorical data
C) It is sensitive to the initial placement of cluster centroids
D) All of the above

Correct Answer[redacted]
Explanation: All of the listed options are disadvantages of the K-means clustering algorithm: it assumes clusters have a spherical shape, it cannot handle categorical data, and it is sensitive to the initial placement of cluster centroids.

How does the K-means clustering algorithm deal with an empty cluster?

A) It reassigns the data points to the nearest non-empty cluster
B) It selects a new centroid for the empty cluster from the remaining data points
C) It removes the empty cluster from the final solution
D) It reinitializes the empty cluster centroid

Correct Answer[redacted]
Explanation: If an empty cluster is encountered in the K-means clustering algorithm, it selects a new centroid for the empty cluster from the remaining data points, typically choosing the data point with the highest distance from its current centroid.

Which of the following is NOT an advantage of K-means clustering?

A) Easy to implement and understand
B) Scalable to large datasets
C) Guaranteed to find the global optimum
D) Converges relatively quickly

Correct Answer:[redacted]
Explanation:K-means clustering is not guaranteed to find the global optimum, as it is sensitive to the initial placement of cluster centroids and can converge to a local minimum.

In K-means clustering, what is the role of the "inertia" or "within-cluster sum of squares"?

A) It measures the similarity between clusters
B) It measures the dissimilarity between clusters
C) It measures the compactness of clusters
D) It measures the separation between clusters

Correct Answe[redacted]

Explanation: In K-means clustering, the "inertia" or "within-cluster sum of squares" measures the compactness of clusters, with lower values indicating tighter clusters.

Which of the following is a limitation of K-means clustering?

A) Sensitivity to the initial placement of cluster centroids
B) Inability to handle missing data
C) Inability to handle categorical data
D) All of the above

Correct Answe[redacted]
Explanation: All of the listed options are limitations of K-means clustering: sensitivity to the initial placement of cluster centroids, inability to handle missing data, and inability to handle categorical data.

What is the primary assumption made by the K-means clustering algorithm?

A) Clusters have a spherical shape
B) Clusters have similar densities
C) Clusters have similar sizes
D) Clusters are linearly separable

Correct Answer:A. Clusters have a spherical shape
Explanation: The primary assumption made by the K-means clustering algorithm is that clusters have a spherical shape, as the algorithm minimizes the Euclidean distance between data points and cluster centroids.

How is the optimal number of clusters typically determined in K-means clustering?

A) By selecting the number of clusters that minimizes the within-cluster variance
B) By using domain knowledge or expert judgment
C) By employing an elbow plot or silhouette analysis
D) By using cross-validation

Correct Answer:
Explanation: The optimal number of clusters in K-means clustering is typically determined by employing an elbow plot or silhouette analysis to identify the point where adding more clusters does not result in a significant improvement in the within-cluster variance.

What is the primary goal of K-means clustering?

A) Dimensionality reduction B) Classification C) Regression D) Partitioning data into clusters

Correct Answer:
Explanation: The primary goal of K-means clustering is to partition data into clusters based on similarity, minimizing the within-cluster variance and maximizing the between-cluster variance.

Which distance metric is most commonly used in K-means clustering?

A) Euclidean distance B) Manhattan distance C) Cosine similarity D)Jaccard similarity

Correct Answer
Explanation: Euclidean distance is the most commonly used distance metric in K-means clustering to measure the similarity between data points.

Naive-Bayes

1. Naïve Bayes classifier algorithms are mainly used in text classification.
a) True b) False

Answer: a
Explanation: Naïve Bayes classifier is a simple probabilistic framework for solving a classification problem. It is used to organize text into categories based on the bayes probability and is used to train data to learn document-class probabilities before classifying text documents.

2. What is the formula for Bayes' theorem? Where (A & B) and (H & E) are events and P(B), P(H) & P(E) ≠ 0.
a) P(H|E) = [P(E|H) * P(E)] / P(H) b) P(A|B) = [P(A|B) * P(A)] / P(B)
c) P(H|E) = [P(H|E) * P(H)] / P(E) d) P(A|B) = [P(B|A) * P(A)] / P(B)

Answer: d
Explanation: Here, P(A) &P(H) is the probability of hypothesis before observing the evidence, P(B) & P(E) is the probability of evidence, P(A|B) & P(H|E) is the posterior probability and P(B|A) & P(E|H) is the likelihood probability.
Since Bayes Theorem states that:
Conditional Probability of A given B = ConditionalprobabilityofBgivenA∗PriorprobabilityofAPriorprobabilityofB

3. Which of the following statement is not true about Naïve Bayes classifier algorithm?
a) It cannot be used for Binary as well as multi-class classifications
b) It is the most popular choice for text classification problems
c) It performs well in Multi-class prediction as compared to other algorithms
d) It is one of the fast and easy machine learning algorithms to predict a class of test datasets

Answer: a
Explanation: Naïve Bayes algorithm can be used for binary as well as multi-class classifications. It is a parametric algorithm, which means it requires a fixed set of assumptions or parameters to simplify the machine's learning process.

4. What is the assumptions of Naïve Bayesian classifier?
a) It assumes that features of a data are completely dependent on each other
b) It assumes that each input variable is dependent and the model is not generative
c) It assumes that each input attributes are independent of each other and the model is generative
d) It assumes that the data dimensions are dependent and the model is generative

Answer: c
Explanation: The assumptions of Naïve Bayes Classifier is that it assumes that each input attributes are independent of each other which is the Naïve part, and the model is generative which is the Bayesian part.

5. Which of the following is not a supervised machine learning algorithm?
a) Decision tree b) SVM for classification problems c) Naïve Bayes d) K-means

Answer: d
Explanation: Decision tree, SVM (Support vector machines) for classification problems and Naïve Bayes are the examples of supervised machine learning algorithm. K-means is an example of unsupervised machine learning algorithm.

6. Which one of the following terms is not used in the Bayes' Theorem?
a) Prior b) Unlikelihood c) Posterior d) Evidence

Answer: b
Explanation: The terms Evidence, Prior, Likelihood and Posterior are used in the Bayes' Theorem. But, the term unlikelihood is not used in the Bayes' Theorem. Bayes Theorem states that Posterior = (Likelihood * Prior) / Evidence.

7. Is the assumption of the Naïve Bayes algorithm a limitation to use it?
a) True b) False

Answer: a
Explanation: It's true that the assumption of the Naïve Bayes' algorithm is a limitation to use it since implicitly it assumes that all the input attributes are mutually independent of each other. But in real life it is almost impossible that we get a set of input attributes which are independent.

8. In which of the following case the Naïve Bayes' algorithm does not work well?
a) When faster prediction is required b) When the Naïve assumption holds true
c) When there is the case of Zero Frequency d) When there is a multiclass prediction

Answer: c

Explanation: When there is a case of "Zero Frequency", the categorical variable is not detected and hence the classifier will not be able to make prediction with an assumption of "Zero" probability.

10. Which one of the following models is a generative model used in machine learning?

a) Linear Regression b) Logistic Regression c) Naïve Bayes d) Support vector machines

Answer: c

Explanation: Naïve Bayes is a type of generative model which is used in machine learning. Linear Regression, Logistic Regression and Support vector machines are the types of discriminative models which are used in machine learning.

12. Identify the parametric machine learning algorithm.

a) CNN (Convolutional neural network) b) KNN (K-Nearest Neighbours)

c) Naïve Bayes d) SVM (Support vector machines)

Answer: c

Explanation: In machine learning, the algorithms which can simplify a function by collecting information about its prediction within a finite set of parameters is defined as parametric machine learning algorithm. Naïve Bayes is a parametric machine learning algorithm whereas CNN, KNN and SVM are non-parametric machine learning algorithms.

13. Which one of the following applications is not an example of Naïve Bayes algorithm?

a) Spam filtering b) Text classification c) Stock market forecasting d) Sentiment analysis

Answer: c Clustering unsupervised

Explanation: Stock market forecasting is one of the most core financial tasks of KNN (K-Nearest Neighbours). Spam filtering, text classification and sentiment analysis is the application of Naïve Bayes algorithm, which uses Bayes theorem of probability for prediction of unknown classes.

14. Arrange the following steps in sequence in order to calculate the probability of an event through Naïve Bayes classifier.

2 **I. Find the likelihood probability with each attribute for each class.**

1 **II. Calculate the prior probability for given class labels.**

3 **III. Put these values in Bayes formula and calculate posterior probability.**

4 **IV. See which class has a higher probability, given the input belongs to the higher probability class.**

a) I → II → III → IV b) II → I → III → IV c) III → II → I → IV d) II → III → I → IV

Answer: b

Explanation: The sequence in which Naïve Bayes calculates the probability of an event is:

II. Calculate the prior probability for given class labels.

I. Find the likelihood probability with each attribute for each class.

III. Put these values in Bayes formula and calculate posterior probability.

IV. See which class has a higher probability, given the input belongs to the higher probability class.

15. "It is easy and fast to predict the class of the test data set by using Naïve Bayes algorithm". Which of the following statement contradicts the above given statement?

a) Because there is no iteration b) Because there is no epoch

c) Because there is an error back propagation

d) Because there are no operations involved in solving a matrix problem

Answer: c

Explanation: Naïve Bayes algorithm is easy and fast to predict the class of the test data set because there is no iteration involved, there is no epoch, there are no operations involved in solving a matrix problem and there is no error back propagation.

1. Which of the following is a key assumption made by the Naive Bayes classifier?

A) Features are independent of each other. B) Features are dependent on each other.
C) The target variable is normally distributed. D) The features are linearly related to the target variable.

Answer: A
Explanation: Naive Bayes assumes that features are conditionally independent, given the class label, which is a simplification that makes the algorithm computationally efficient.

2. In Naive Bayes, which distribution is commonly assumed for continuous data?

A) Poisson distribution B) Exponential distribution C) Normal distribution D) Binomial distribution

Answer: C
Explanation: For continuous data, Naive Bayes typically assumes that the features follow a normal (Gaussian) distribution.

3. What is the main advantage of the Naive Bayes classifier?

A) It performs well even with a small amount of training data.
B) It can only be used for binary classification tasks.
C) It is sensitive to irrelevant features.
D) It requires a lot of computational resources.

Answer: A
Explanation: Naive Bayes is a probabilistic classifier that tends to perform well with relatively small datasets, especially when the features are conditionally independent.

4. In Naive Bayes, what is the role of Bayes' theorem?

A) It helps in calculating the probability of a feature belonging to a particular class.
B) It is used to optimize the hyperparameters of the model.
C) It is used for feature selection.
D) It calculates the loss function for model training.

Answer: A)
Explanation: Bayes' theorem is used to calculate the posterior probability of a class given the features, which is the core of the Naive Bayes algorithm.

5. Which type of data can Naive Bayes be applied to?

A) Only numerical data B) Only categorical data
C) Both categorical and numerical data D) Only data with binary features

Answer: C
Explanation: Naive Bayes can be applied to both types of data. For categorical data, it uses the multinomial distribution, and for continuous data, it assumes a normal distribution.

6. What is the primary disadvantage of the Naive Bayes classifier?

A) It requires a large amount of training data.
B) It assumes that features are independent, which is often unrealistic.

C) It is computationally expensive.
D) It performs poorly with large datasets.

Answer: B
Explanation: The independence assumption is a strong simplification that often does not hold in real-world data, which can lead to suboptimal performance.

7. In the context of Naive Bayes, what does the term "likelihood" refer to?

A) The probability of a class given the features
B) The probability of the features given the class
C) The prior probability of a class
D) The probability of the features occurring

Answer: B
Explanation: In Naive Bayes, the likelihood refers to the probability of observing the given features under each class.

9. Naive Bayes is particularly suited for which of the following tasks?
A) Regression with large datasets
B) Multiclass classification problems
C) Feature engineering for deep learning models
D) Clustering similar data points

Answer: B
Explanation: Naive Bayes is particularly effective for multiclass classification, where it can efficiently calculate probabilities for multiple classes.

10. Which of the following would likely reduce the performance of a Naive Bayes classifier?
A) Using features that are highly correlated
B) Using a smaller training dataset
C) Using a very large number of features
D) Using Laplace smoothing
Answer: A) Using features that are highly correlated

Questions

MCQ Questions & Answers

What is Artificial intelligence?

A. Putting your intelligence into Computer
B. Programming with your own intelligence
C. Making a Machine intelligent
D. Playing a Game

ANSWER: C

2 Strong Artificial Intelligence is
A. the embodiment of human intellectual capabilities within a computer
B. a set of computer programs that produce output that would be considered to reflect intelligence if it were generated by humans
C. the study of mental faculties through the use of mental models implemented on a computer
D. all of the mentioned

ANSWER: A

3 In which of the following situations might a blind search be acceptable?

A. real-life situation B. complex game C. small search space D. all of the mentioned

ANSWER: C

4 Which search method takes less memory?

A. Depth-First Search B. Breadth-First search C. Optimal search D. Linear Search

ANSWER: A

5 A heuristic is a way of trying

A. To discover something or an idea embedded in a program
B. To search and measure how far a node in a search tree seems to be from a goal
C. To compare two nodes in a search tree to see if one is better than the other is
D. All of the mentioned

ANSWER: D

6 Which is not a property of representation of knowledge?

A. Representational Verification B. Representational Adequacy
C. Inferential Adequacy D. Inferential Efficiency

ANSWER: A

7. turing developed a technique for determining whether a computer could or could not demonstrate the artificial Intelligence, Presently, this technique is called

A. Turing Test B. Algorithm C. Boolean Algebra D. Logarithm

ANSWER: A

8 A Personal Consultant knowledge base contain information in the form of

A. parameters B. contexts C. production rules D. all of the mentioned

ANSWER: D

9 Which approach to speech recognition avoids the problem caused by the variation in speech patterns among different speakers?

A. Continuous speech recognition B. Isolated word recognition
C. Connected word recognition D. Speaker-dependent recognition

ANSWER: D

10 Which of the following, is a component of an expert system?

A. inference engine B. knowledge base C. user interface D. all of the mentioned

ANSWER: D

11 A computer vision technique that relies on image templates is

A. edge detection B. binocular vision C. model-based vision D. robot vision

ANSWER: C

12 the agency that has funded a great deal of American Artificial Intelligence research, is part of the Department of

A. Defense B. Energy C. Education D. Justice

ANSWER: A

13 Which of these schools was not among the early leaders in Artificial Intelligence research?
A. Dartmouth University B. Harvard University
C. Massachusetts Institute of Technology D. Stanford University

ANSWER: B

14 Who is the "father" of artificial intelligence?
A. Fisher Ada B. John McCarthy C. Allen Newell D. Alan Turning

ANSWER: B

15 A process that is repeated, evaluated, and refined is called
A. diagnostic B. descriptive C. interpretive D. iterative

ANSWER: D

16 Visual clues that are helpful in computer vision include
A. color and motion B. depth and texture C. height and weight D. color and motion, depth and texture

ANSWER: D

17 General games involves
A. Single-agent B. Multi-agent
C. Neither Single-agent nor Multi-agent D. Only Single-agent and Multi-agent

ANSWER: D

18 Adversarial search problems uses
A. Competitive Environment B. Cooperative Environment
C. Neither Competitive nor Cooperative Environment
D. Only Competitive and Cooperative Environment

ANSWER: A

19 Zero sum game has to be a game.
A. Single player B. Two player C. Multiplayer D. Three player

ANSWER: C

20 A game can be formally defined as a kind of search problem with the following components.
A. Initial State B. Successor Function C. Terminal Test D. All of the mentioned

ANSWER: D

21 The initial state and the legal moves for each side define the for the game.
A. Search Tree B. Game Tree C. State Space Search D. Forest

ANSWER: B

22 General algorithm applied on game tree for making decision of win/lose is
A. DFS/BFS Search Algorithms B. Heuristic Search Algorithms
C. Greedy Search Algorithms D. MIN/MAX Algorithms

ANSWER: D

23 What is the complexity of minirnax algorithm?
A. Same as of DES　　B. Space — bm and time — bm
C. Time — bm and space — bm　　D. Same as BFS

ANSWER: A

24 Which is the most straightforward approach for planning algorithm?
A. Best-first search　　B. State-space search　　C. Depth-first search　　D. Hill-climbing search

ANSWER: B

25 What are taken into account of state-space search?
A. Postconditions　　B. Preconditions　　C. Effects　　D. Both Preconditions & Effects

ANSWER: D

26 How many ways are available to solve the state-space search?
A. 1　　B. 2　　C. 3　　D. 4

ANSWER: B

27 What is the other name for forward state-space search?
A. Progression planning　　B. Regression planning　　C. Test planning　　D. None of the mentioned

ANSWER: A

28 How many states are available in state-space search?
A. 1　　B. 2　　C. 3　　D. 4

ANSWER: D

29 What is the main advantage of backward state-space search?
A. Cost　　B. Actions　　C. Relevant actions　　D. All of the mentioned

ANSWER: C

30 What is the other name of the backward state-space search?
A. Regression planning　　B. Progression planning　　C. State planning　　D. Test planning

ANSWER: A

31 What is meant by consistent in state-space search?
A. Change in the desired literals　　B. Not any change in the literals
C. No change in goal state　　D. None of the mentioned

ANSWER: B

32 What will happen if a predecessor description is generated that is satisfied by the initial state of the planning problem?
A. Success　　B. Error　　C. Compilation　　D. Termination

ANSWER: 0

33 Which approach is to pretend that a pure divide and conquer algorithm will work?
A. Goal independence　　B. Subgoal independence
C. Both Goal & Subgoal independence　　D. None of the mentioned

ANSWER: B

34 Which search is equal to minimax search but eliminates the branches that can't influence the final decision?

A. Depth-first search B. Breadth-first search C. Alpha-beta pruning D. None of the mentioned

ANSWER: C

35 Which values are independant in minimax search algorithm?

A. Pruned leaves x and y B. Every states are dependant

C. Root is independant D. None of the mentioned

ANSWER: A

36 To which depth does the alpha-beta pruning can be applied?

A. 10 states B. 8 States C. 6 States D. Any depth

ANSWER: D

37 Which search is similar to minimax search?

A. Hill-climbing search B. Depth-first search C. Breadth-first search D. All of the mentioned

ANSWER: B

38 Which value is assigned to alpha and beta in the alpha-beta pruning?

A. Alpha = max B. Beta = min C. Beta = max D. Both Alpha = max & Beta = min

ANSWER: D

39 Where does the values of alpha-beta search get updated?

A. Along the path of search B. Initial state itself C. At the end D. None of the mentioned

ANSWER: A

40 How the effectiveness of the alpha-beta pruning gets increased?

A. Depends on the nodes B. Depends on the order in which they are executed

C. All of the mentioned D. None of the mentioned

ANSWER: A

43 Which function is used to calculate the feasibility of whole game tree?

A. Evaluation function B. Transposition C. Alpha-beta pruning D. All of the mentioned

ANSWER: A

44 Knowledge and reasoning also play a crucial role in dealing with environment.

A. Completely Observable B. Partially Observable

C. Neither Completely nor Partially Observable D. Only Completely and Partially Observable

ANSWER: B

45 Treatment chosen by doctor for a patient for a disease is based on

A. Only current symptoms B. Current symptoms plus some knowledge from the textbooks

C. Current symptoms plus some knowledge from the textbooks plus experience

D. All of the mentioned

ANSWER: C

46 Choose the correct option.
A. Knowledge base (KB. is consists of set of statements.
B. Inference is deriving a new sentence from the KB. A. A is true, B is true B. A is false, B is false
C. A is true, B is false D. A is false, B is true

ANSWER: A

49 Which is not a property of representation of knowledge?
A. Representational Verification B. Representational Adequacy
C. Inferential Adequacy D. Inferential Efficiency

ANSWER: A

68 How many terms are required for building a bayes model?
A. 1 B. 2 C. 3 D. 4

ANSWER: C

70 Where does the bayes rule can be used?
A. Solving queries B. Increasing complexity C. Decreasing complexity D. ANSWERing probabilistic query

ANSWER: D

71 What does the bayesian network provides?
A. Complete description of the domain B. Partial description of the domain
C. Complete description of the problem D. None of the mentioned

ANSWER: A

Which of the following is true for neural networks?
0) The training time depends on the size of the network.
(ii) Neural networks can be simulated on a conventional computer.
(iii) Artificial neurons are identical in operation to biological ones

A. All of the mentioned B. (ii) is true
C. (i) and (ii) are true D. None of the mentioned

ANSWER: [redacted]

What is the goal of artificial intelligence?
A. To solve real-world problems B. To solve artificial problems
C. To explain various sorts of intelligence D. To extract scientific causes

ANSWER: C

An algorithm is complete if
A. It terminates with a solution when one exists B) It starts with a solution 8th IT- AI Question Bank
C. It does not terminate with a solution D. It has a loop

ANSWER: A

Which is true regarding BFS (Breadth First Search)?
A. BFS will get trapped exploring a single path

B. The entire tree so far been generated must be stored in BFS
C. BFS is not guaranteed to find a solution if exists
D. BFS is nothing but Binary First Search

ANSWER: B

What is a heuristic function?
A. A function to solve mathematical problems
B. A function which takes parameters of type string and returns an integer value
C. A function whose return type is nothing
D. A function that maps from problem state descriptions to measures of desirability

ANSWER: D

What was originally called the “imitation game” by its creator?
A The logic theorist B Lisp C The turing test

Answer: C

What is the term used for describing the judgmental or commonsense part of problem solving?
A Analytical B Heuristic

ANSWER : B

The characteristics of the computer system capable of thinking, reasoning and learning is known is
A Machine intelligence B Artificial intelligence C Human intelligence

ANSWER : B

Artificial intelligence has its expansion in the following application.
A Planning and scheduling B Robotics C All of the above

ANSWER : C

A problem in a search space is defined by one of these state
A Initial state B Last state C Intermediate state D All of the mentioned

ANSWER : C

Rational agent is the one who always does the right thing.
A True B False

ANSWER: A

Performance measures are fixed for all agents.
A True B False

ANSWER: A

Which element in the agent are used for selecting external actions?
A Perceive B Performance C Learning D Actuator

ANSWER: B

LAST YEAR EXAM

Question 1

2. Given the Arad/Bucharest route finding problem and the heuristics table below, show the list of visited nodes using the following search techniques:

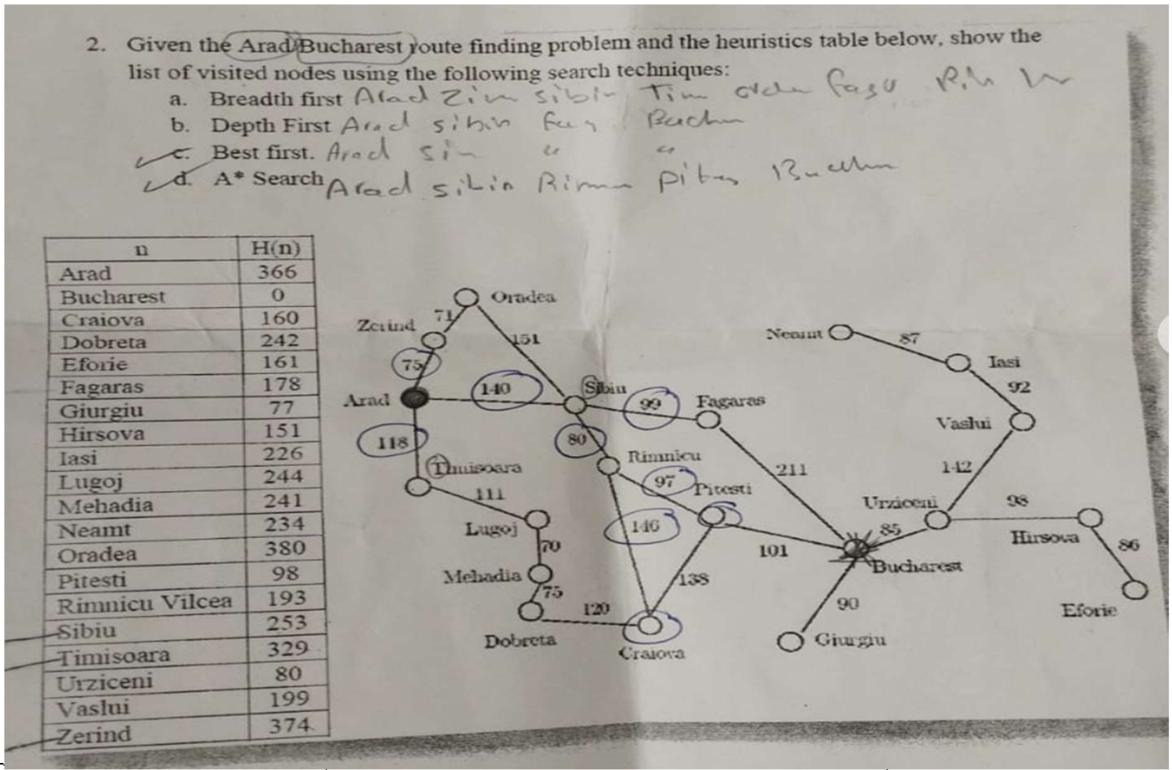
a. Breadth first

b. Depth First

c. Best first

d. A* Search

Heuristics Table:



2. Given the Arad/Bucharest route finding problem and the heuristics table below, show the list of visited nodes using the following search techniques:
   a. Breadth first
   b. Depth First
   c. Best first.
   d. A* Search

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

n	H(n)
Arad	366
Bucharest	0
Craiova	160
Dobreta	242
Eforie	161
Fagaras	178
Giurgiu	77
Hirsova	151
Iasi	226
Lugoj	244
Mehadia	241
Neamt	234
Oradea	380
Pitesti	98
Rimnicu Vilcea	193
Sibiu	253
Timisoara	329
Urziceni	80
Vaslui	199
Zerind	374

Question 2

1. Compare briefly between Systems that "Act / Behave Rationally", and Systems that "Act / Behave Humanly".

2. Describe briefly Intelligent Agents.

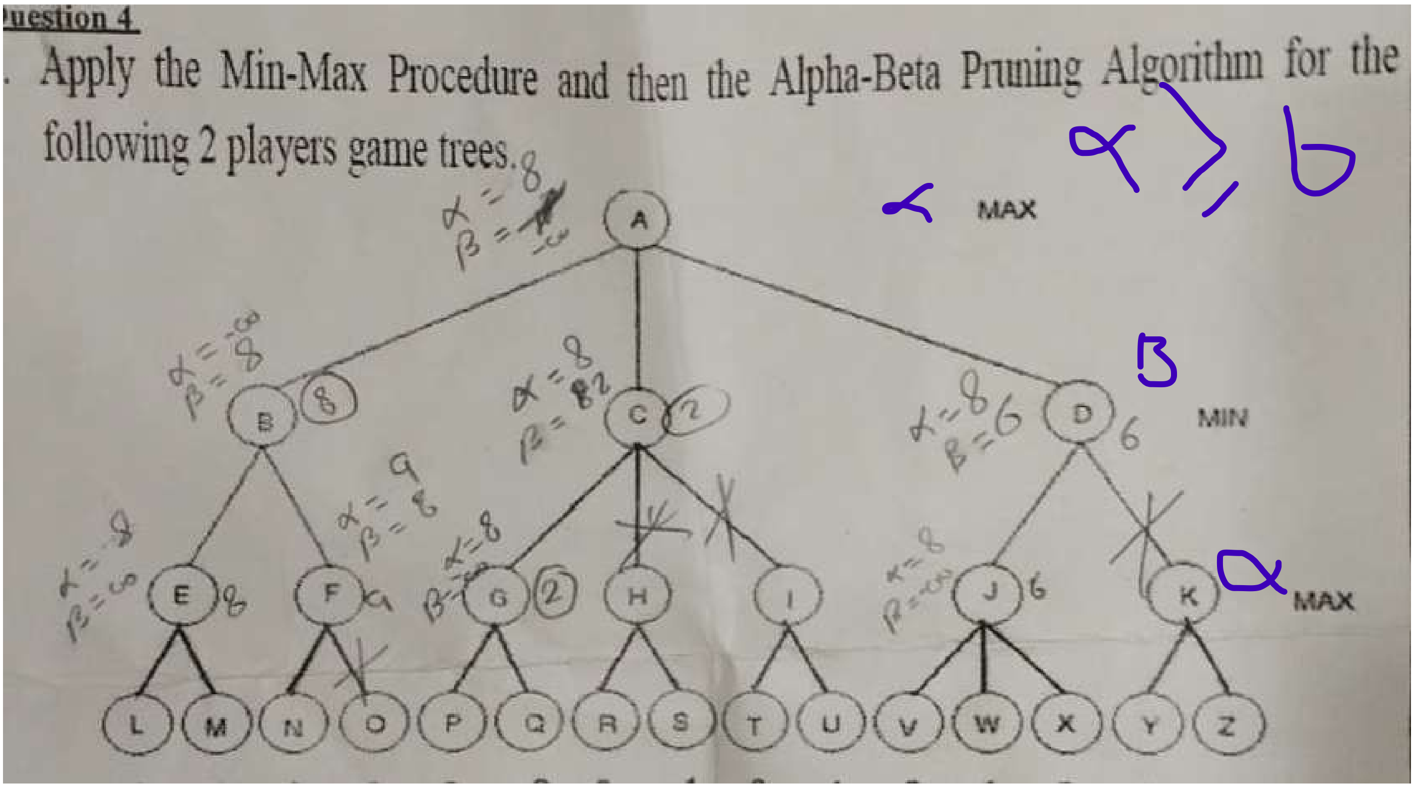
3. Discuss PEAS, and how is it employed to specify a task environment.Question 3

Apply the k-means algorithm to cluster the following patterns into 3 partitions using the Manhattan distance measure based on selecting P1, P6 and P7 as initial centroids. Show the first iteration of the clustering process and compute the new centroids for the next iteration.

	P1	**P2**	**P3**	**P4**	**P5**	**P6**	**P7**	**P8**	**P9**
	1	5	9	8	2	5	5	7	3
	1	7	1	1	1	6	5	1	1
	1	2	3	3	1	2	2	3	1

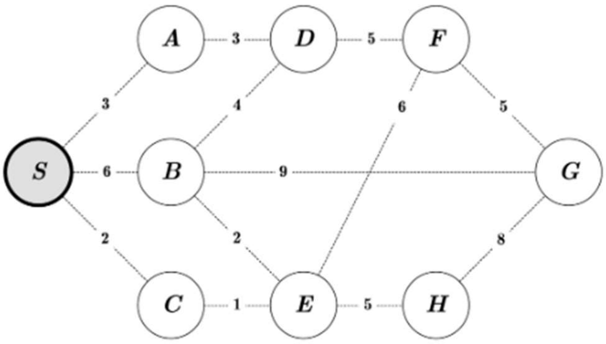
Question 4

1. Apply the Min-Max Procedure and then the Alpha-Beta Pruning Algorithm for the following 2 players game trees.



LECS EXAMPLES

Exercise: UCS



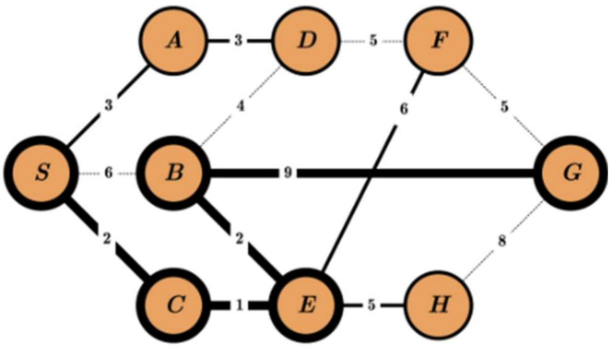
Priority Queue:

S_0

Order of Visit:

ANS

Exercise: UCS



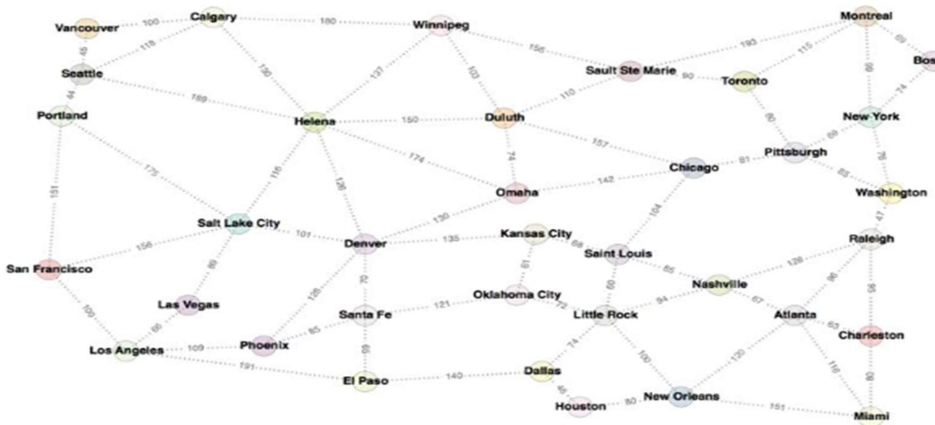
Priority Queue:

S_0	C_2	A_3	E_3	B_5	D_6	H_8	F_9	G_{14}

Order of Visit:

S C A E B D H F G

Informed search



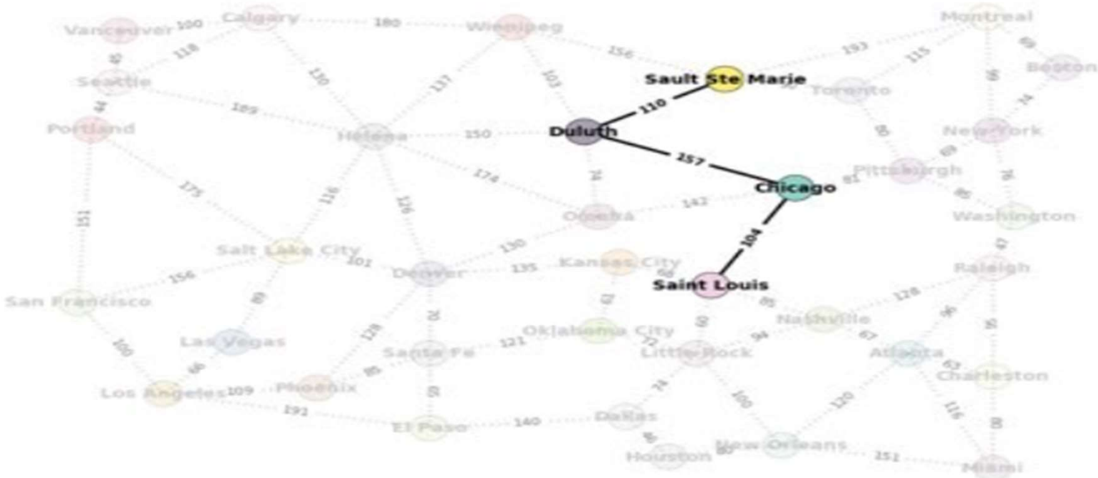
Atlanta	272
Boston	240
Calgary	334
Charleston	322
Chicago	107
Dallas	303
Denver	270
Duluth	110
El Paso	370
Helena	254
Houston	332
Kansas City	176
Las Vegas	418
Little Rock	240
Los Angeles	484
Miami	389
Montreal	193
Nashville	221
New Orleans	322
New York	195
Oklahoma City	237
Omaha	150
Phoenix	396
Pittsburgh	152
Portland	452
Raleigh	251
Saint Louis	180
Salt Lake City	344
San Francisco	499
Santa Fe	318
Sault Ste Marie	0
Seattle	434
Toronto	90
Vancouver	432
Washington	238
Winnipeg	156

Heuristic!

The distance is the straight line distance. The goal is to get to Sault Ste Marie, so all the distances are from each city to Sault Ste Marie.

Examples using the map

Start: Saint Louis
Goal: Sault Ste Marie

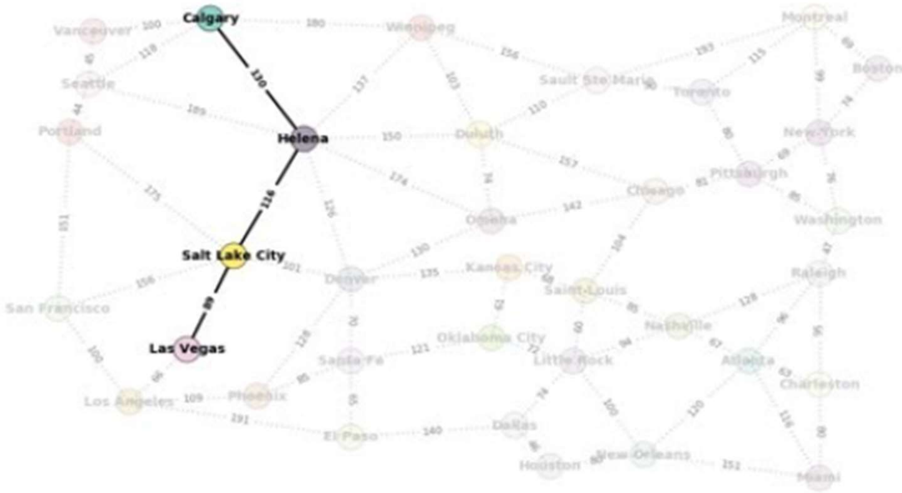


Greedy search

Examples using the map

Start: Las Vegas

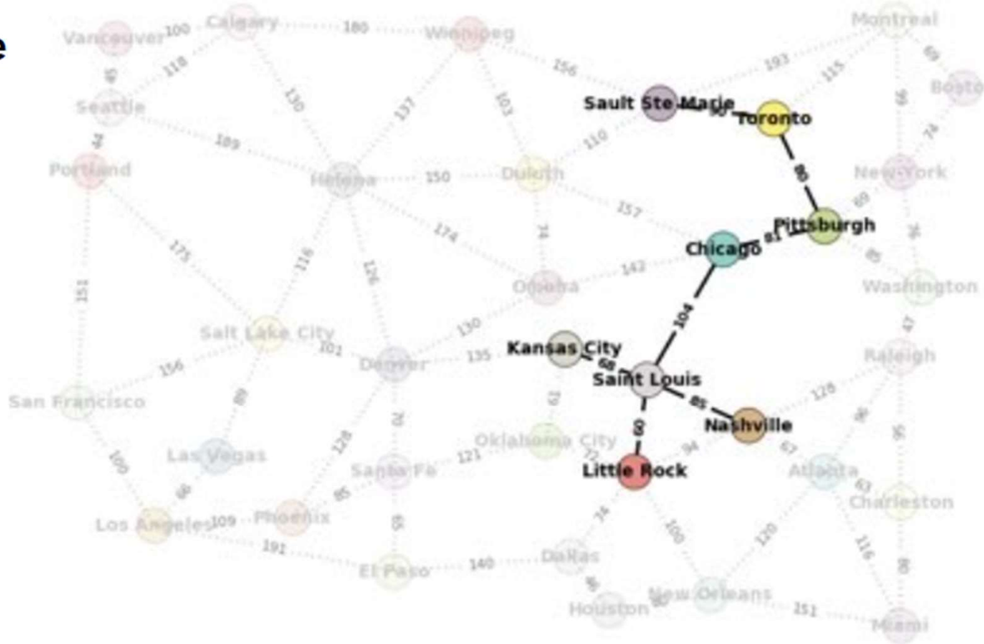
Goal: Calgary



Examples using the map

Start: Saint Louis

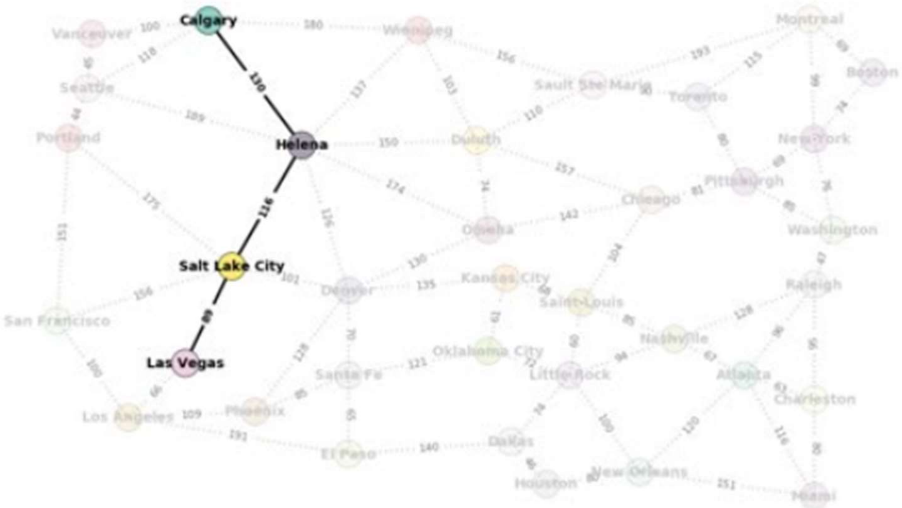
Goal: Sault Ste Marie



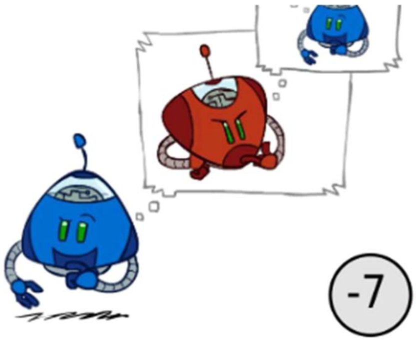
Examples using the map

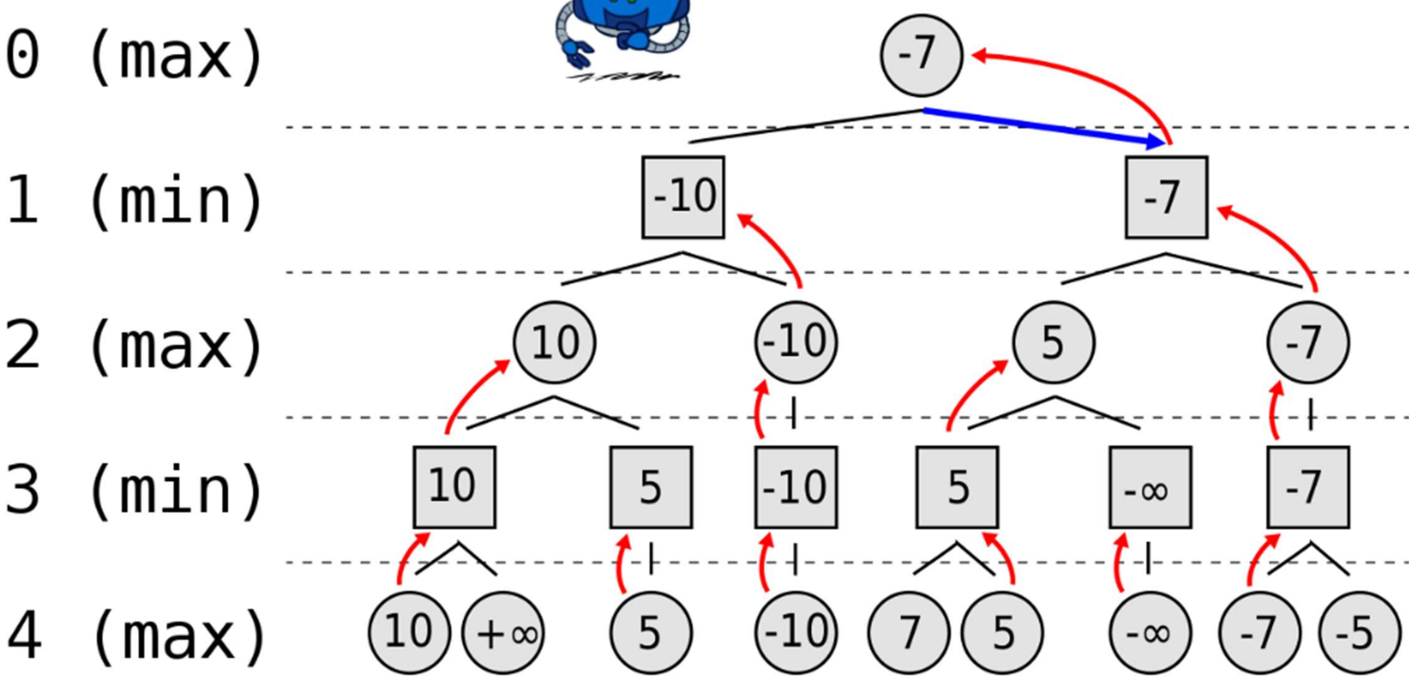
Start: Las Vegas

Goal: Calgary

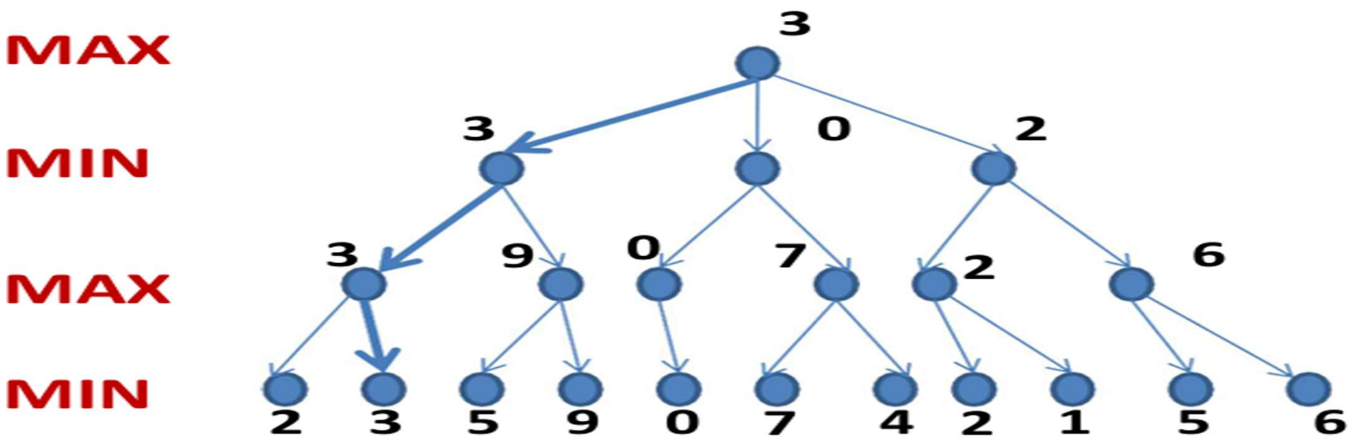


Minimax Tree Example

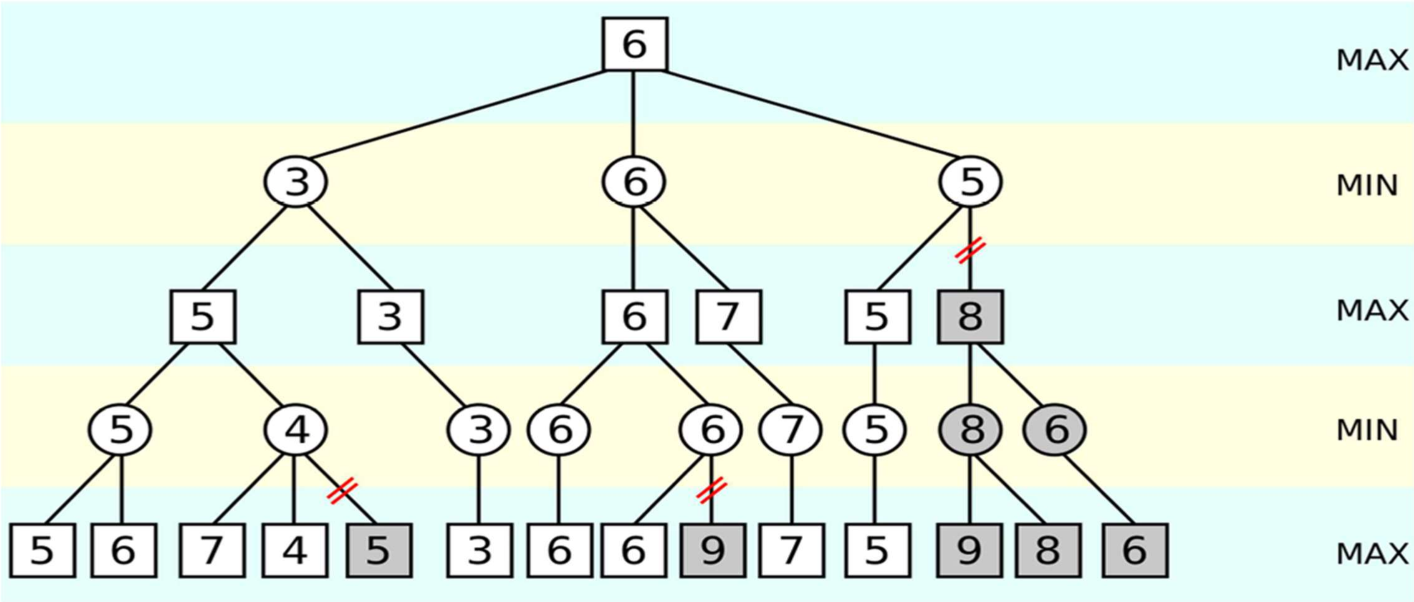




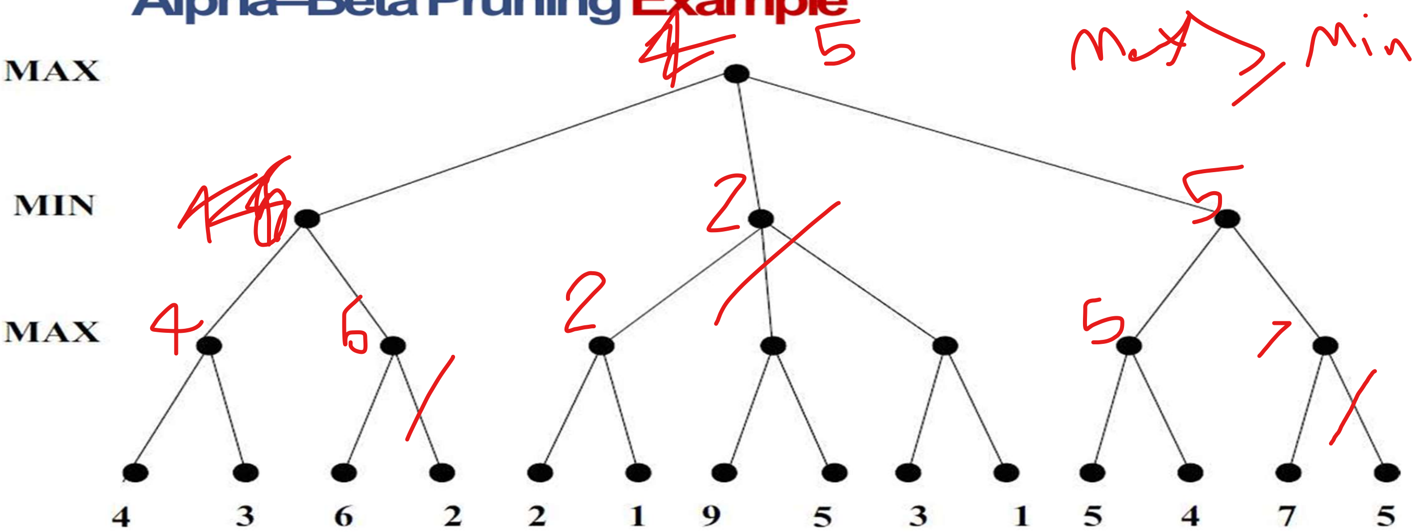
Minimax to a Fixed Ply Depth



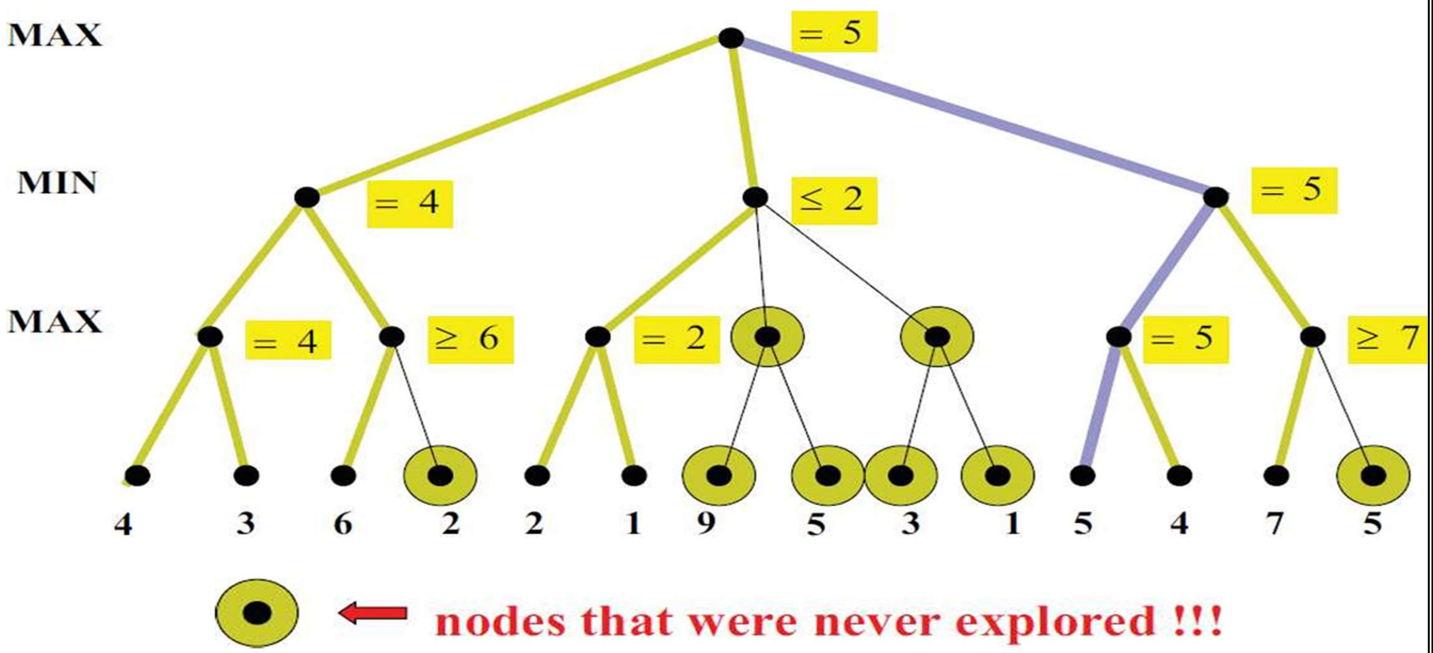
Alpha–Beta Pruning Procedure..An Illustration



Alpha–Beta Pruning Example



Alpha–Beta Pruning Example



- Given: {2,4,10,12,3,20,30,11,25}, k=2

- Stop as the clusters with these means are the same.

- Given: {2,4,10,12,3,20,30,11,25}, k=2
    1. Randomly assign two centers: $C_1=3$, $C_2=4$
    2. $K_1=\{2,3\}$, $K_2=\{4,10,12,20,30,11,25\}$, $C_1=2.5$, $C_2=16$
    3. $K_1=\{2,3,4\}$, $K_2=\{10,12,20,30,11,25\}$, $C_1=3$, $C_2=18$
    4. $K_1=\{2,3,4,10\}$, $K_2=\{12,20,30,11,25\}$, $C_1=4.75$, $C_2=19.6$
    5. $K_1=\{2,3,4,10,11,12\}$, $K_2=\{20,30,25\}$, $C_1=7$, $C_2=25$
    6. $K_1=\{2,3,4,10,11,12\}$, $K_2=\{20,30,25\}$, $C_1=7$, $C_2=25$
- Stop as the clusters with these means are the same.

Cluster the following dataset, 8 points, into three clusters using K-means based on Manhattan distance.

{(2, 10), (2, 5), (8, 4), (5, 8), (7, 5), (6, 4), (1, 2), (4, 9)}.

First Iteration:

Point		Centers			Cluster (C?)
		C1 (2, 10)	C2 (5, 8)	C3 (1, 2)	
		Dist from C1	**Dist from C2**	**Dist from C3**	
A1	(2, 10)	|2-2|+|10-10|=0	|2-5|+|10-8|=5	|2-1|+|10-2|=9	1
A2	(2, 5)	5	6	4	3
A3	(8, 4)	12	7	9	2
A4	(5, 8)	5	0	10	2
A5	(7, 5)	10	5	9	2
A6	(6, 4)	10	5	7	2
A7	(1, 2)	9	10	0	3
A8	(4, 9)	3	2	10	2

Second Iteration:

Point		Centers			Cluster (C?)
		C1(2, 10)	C2(6, 6)	C3(1.5, 3.5)	
		Dist from C1	**Dist from C2**	**Dist from C3**	
A1	(2, 10)	0	8	7	1
A2	(2, 5)	5	5	2	3
A3	(8, 4)	12	4	7	2
A4	(5, 8)	5	3	8	2
A5	(7, 5)	10	3	7	2
A6	(6, 4)	10	4	5	2
A7	(1, 2)	9	9	2	3
A8	(4, 9)	3	5	8	1

Example 1: Given The Following Dataset, Cluster it using the K-Mea
(3,4) , (3,6) , (3,8) , (4,5) , (4,7) , (5,1) , (5,5) , (7,3) , (7,5) , (8,5)
Set K=2 and the Initial Centroids m1(3,4), m2(8,5) , Use Euclidian Distance

1st Iteration

Example 1: Given The Following Dataset, Cluster it using the K-Mea[ns]
(3,4) , (3,6) , (3,8) , (4,5) , (4,7) , (5,1) , (5,5) , (7,3) , (7,5) , (8,5)
Set K=2 and the Initial Centroids m1(3,4), m2(8,5) . Use Euclidian Distance

1st Iteration

Point	M1 Distance	M2 Distance	Cluster
(3,4)	$\sqrt{(3-3)^2+(4-4)^2} = 0.00$	$\sqrt{(3-8)^2+(4-5)^2} = 5.10$	Cluster1
(3,6)	$\sqrt{(3-3)^2+(6-4)^2} = 2.00$	$\sqrt{(3-8)^2+(6-5)^2} = 5.10$	Cluster1
(3,8)	$\sqrt{(3-3)^2+(8-4)^2} = 4.00$	$\sqrt{(3-8)^2+(8-5)^2} = 5.83$	Cluster1
(4,5)	$\sqrt{(4-3)^2+(5-4)^2} = 1.41$	$\sqrt{(4-8)^2+(5-5)^2} = 4.00$	Cluster1
(4,7)	$\sqrt{(4-3)^2+(7-4)^2} = 3.16$	$\sqrt{(4-8)^2+(7-5)^2} = 4.47$	Cluster1
(5,1)	$\sqrt{(5-3)^2+(1-4)^2} = 3.61$	$\sqrt{(5-8)^2+(1-5)^2} = 5.00$	Cluster1
(5,5)	$\sqrt{(5-3)^2+(5-4)^2} = 2.24$	$\sqrt{(5-8)^2+(5-5)^2} = 3.00$	Cluster1
(7,3)	$\sqrt{(7-3)^2+(3-4)^2} = 4.12$	$\sqrt{(7-8)^2+(3-5)^2} = 2.24$	Cluster2
(7,5)	$\sqrt{(7-3)^2+(5-4)^2} = 4.12$	$\sqrt{(7-8)^2+(5-5)^2} = 1.00$	Cluster2
(8,5)	$\sqrt{(8-3)^2+(5-4)^2} = 5.10$	$\sqrt{(8-8)^2+(5-5)^2} = 0.00$	Cluster2

- Cluster the following dataset , into two clusters using K-means based on Manhattan distance.

Given :{2,4,5,10,12,13}, K=2

- 1. Randomly assign two centers : C1 = 4 , C2=12.
- 2. k1={2,4,5} , k2={10,12,13}, C1 = 3.67 , C2=11.67.
- 3. k1={2,4,5} , k2={10,12,13}, C1 = 3.67 , C2=11.67.
- Stop as the clusters with these means are the same.

Cluster the following dataset , 4 points into two clusters using K-means based on Manhattan distance.

Given :{(2,3),(3,4),(10,10),(11,11)}.

First iteration :

Initial random centers are : C1= (2,3) , C2= (10,10).

	Centers		
Points	C1= (2,3)	C2= (10,10).	cluster
A1 (2,3)	|2-2|+|3-3|=0	|2-10|+|3-10|=15	1
A2 (3,4)	|3-2|+|4-3|=2	|3-10|+|4-10|=13	1
A3 (10,10)	|10-2|+|10-3|=15	|10-10|+|10-10|=0	2
A4 (11,11)	|11-2|+|11-3|=17	|11-10|+|11-10|=2	2

Now, calculate the mean of each cluster for the next iteration .

- **Second iteration :**
- centers are : C1= (2.5,3.5) , C2= (10.5,10.5).

Centers			
Points	C1= (2.5,3.5)	C2= (10,.510.5).	cluster
A1 (2,3)	|2-2.5|+|3-3.5|=0	|2-10.5|+|3-10.5|=15	1
A2 (3,4)	|3-2.5|+|4-3.5|=2	|3-10.5|+|4-10.5|=13	1
A3 (10,10)	|10-2.5|+|10-3.5|=15	|10-10.5|+|10-10.5|=0	2
A4 (11,11)	|11-2.5|+|11-3.5|=17	|11-10.5|+|11-10.5|=2	2

- Now, calculate the mean of each cluster for the next iteration .

- The new centers are equal to the previous centers then will stop at

K1= { (2,3) , (3,4) }, K2= { (10,10) , (11,11) }.

Naïve Bayes Classifier: Training Dataset

Class:
C1:buys_computer = 'yes'
C2:buys_computer = 'no'

Data to be classified:
X = (age <=30,
Income = medium,
Student = yes
Credit_rating = Fair)

age	income	student	credit_ rating	comp uter
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

$P(C_i)$: P(buys_computer = "yes") = 9/14 = 0.643

P(buys_computer = "no") = 5/14= 0.357

Compute $P(X|C_i)$ for each class

P(age = "<=30" | buys_computer = "yes") = 2/9 = 0.222

P(age = "<= 30" | buys_computer = "no") = 3/5 = 0.6

P(income = "medium" | buys_computer = "yes") = 4/9 = 0.444

P(income = "medium" | buys_computer = "no") = 2/5 = 0.4

P(student = "yes" | buys_computer = "yes) = 6/9 = 0.667

P(student = "yes" | buys_computer = "no") = 1/5 = 0.2

P(credit_rating = "fair" | buys_computer = "yes") = 6/9 = 0.667

P(credit_rating = "fair" | buys_computer = "no") = 2/5 = 0.4

- **X = (age <= 30 , income = medium, student = yes, credit_rating = fair)**

$P(X|C_i)$: P(X|buys_computer = "yes") = 0.222 x 0.444 x 0.667 x 0.667 = 0.044

P(X|buys_computer = "no") = 0.6 x 0.4 x 0.2 x 0.4 = 0.019

$P(X|C_i)*P(C_i)$: P(X|buys_computer = "yes") * P(buys_computer = "yes") = 0.028

P(X|buys_computer = "no") * P(buys_computer = "no") = 0.007

Therefore, X belongs to class ("buys_computer = yes")

Example of Confusion Matrix:

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	**6954**	**46**	7000
buy_computer = no	**412**	**2588**	3000
Total	7366	2634	10000

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%)
cancer = yes	**90**	**210**	300	30.00 (*sensitivity*
cancer = no	**140**	**9560**	9700	98.56 (*specificity)*
Total	230	9770	10000	96.40 (*accuracy*)

- *Precision* = 90/230 = 39.13% *Recall* = 90/300 = 30.00%

Actual Class\Predicted class	cancer = yes	cancer = no	Total	Recognition(%
cancer = yes	90	210	300	30.00 (*sensitivit*
cancer = no	140	9560	9700	98.56 (*specificit*
Total	230	9770	10000	96.40 (*accurac*

Precision = 90/230 = 39.13%

Recall = 90/300 = 30.00%

F-Score = 2*P*R/P+R = 2* 39.13*30/(39.13+30)= 33.96

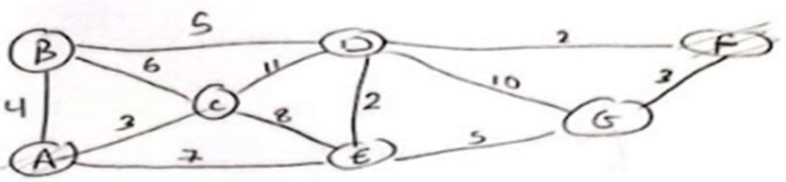
Accuracy = (90+ 9560)/10000

Actual class\Predicted class	buy_computer = yes	buy_computer = no	Total
buy_computer = yes	6954	46	7000
buy_computer = no	412	2588	3000
Total	7366	2634	10000

ccuracy: Accuracy = (TP + TN)/All = (6954 + 2588)/10000

Precision = (TP)/ (TP+ FP) = 6954/(7366) =

Recall= TP/(TP+ FN)= 6954/7000 =



A-f

Example ③

حل

فقط USC

Bfs

O.V: A, B, C, E, D, G, f

f.P: A B D f

Queue

A (~~B~~, C, E)

AB (~~E~~, E, D)

AC (~~E~~, D)

AE (~~D~~, G)

ABD (~~E~~, f)

AEG (f)

$\boxed{ABDf}$ goal

Dfs

O.V A, B, D, f

f.P A, B, D, f

Stack

A (~~B~~, C, E)

AB (~~D~~, C, E)

ABD (~~f~~, G, C, E)

$\boxed{ABDf}$ goal

UCS

O.V A, C, B, E, D, f

f.P A B D F

Cost 11

Queue

fn = gn = Σ lowest cost

A - B.4

A - C = 3

A - E = 7

AC - B = 3 + 6 = 9 X

AC - D = 3 + 11 = 14 X

AC - E = 3 + 8 = 11 X

AB - D = 4 + 5 = 9

AE - D = 7 + 2 = 9 X

AE - G = 7 + 5 = 13

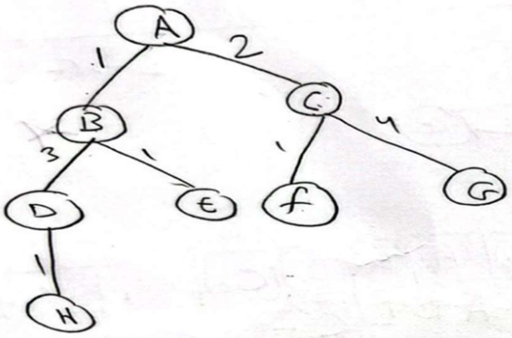
ABD - G = 9 + 10 = 19

ABD - f = 9 + 2 = 11

ABDf = 11 goal

Ex①

A -> f



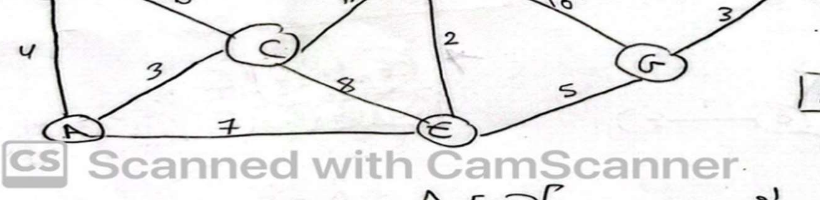
Queue

A_0	B_1	C_2	D_4	E_2	f_3	G_6

A B C E_2 f

ACf

A-f

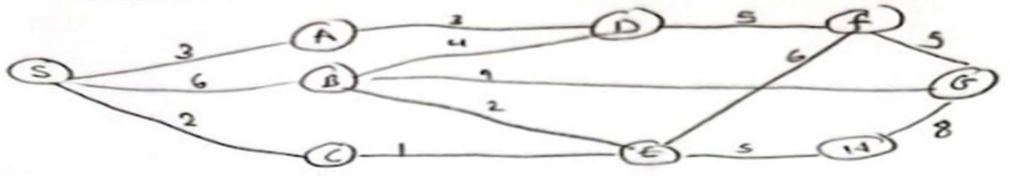


A_0	C_3	B_4	E_7	D_{14}	D_9	~~X~~	f_{11}	G_{19}

A C B E D f_{11}

A E D f أو $\boxed{ABDf}$ (11)

Example ①



$\boxed{S \to G}$

BFS

ordered visited: S, A, B, C, D, E, G

final path: SBG

Queue Fifo

S (A, B, C)

SA (B, C, D)

SB (E, D, E, G)

SC (D, E, G)

SAD (E, G, f)

SBE (G, f, H)

SBG goal

DFS

O.V SAD, f, E, H, G

fp SADfEHG

Stack

S (A, B, C)

SA (D, B, C)

SAD (f, B, C)

SADf (E, G, B, C)

SADfE (H, G, B, C)

SADfEH (G, B, C)

SADfG goal

UCS

O.V : S, C, A, E, B, D, H, f, G

fp : SCEBG

Cost : 14

g(n) = Σ lowest cost

S — S − A = 3 ✓
S − B = 6 ✗
S − C = 2 ✓

SC — SC − E = 2 + 1 = 3 ✓

SA — SA − D = 3 + 3 = 6 ✓

SCE — SCE − B = 3 + 2 = 5 ✓
SCE − f = 3 + 6 = 9 ✓
SCE − H = 3 + 5 = 8 ✓

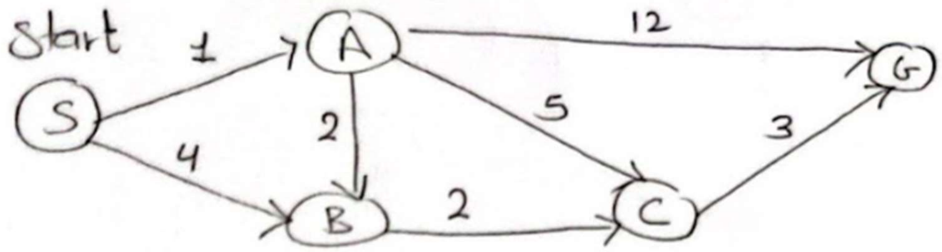
SCEB — SCEB − D = 5 + 4 = 9 ✗
SCEB − G = 5 + 9 = 14 ✓

SAD — SAD − f = 6 + 5 = 11 ✗

SCEH — SCEH − G = 8 + 8 = 16

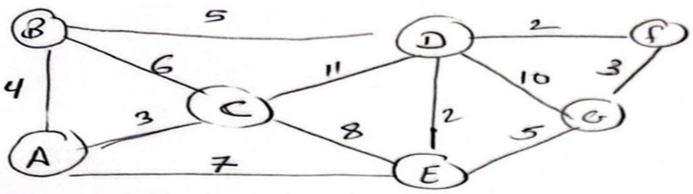
SCEf — SCEf − G = 9 + 8 = 17

SCEBG = 14 goal



Time

n	h(n)
S	7
A	6
B	4
C	2
G	0



A-F

Distance

n	h(n)
A	28
B	30
C	56
D	60
E	44
F	0
G	36

Department : Management Information Systems
Year : 4th Year
Course Title : Artificial Intelligence
Exam : MidTerm
Lecturer : Dr. Abeer A. Amer
Semester : Second
Date : 25/03/2024
Time Allowed : One Hour

Question .1. True / False:

1. The game of Poker is a single agent. ×
2. A solution to a problem is a path from the initial state to a goal state. Solution quality is measured by the path cost function, and an optimal solution has the highest path cost among all solutions. ×
3. likes(jin, jax). the same as likes(jax, jin). ×
4. food(b, c, d(a)). is ?-food(X, Y). a true query? ×
5. If sky is blue, everyone likes it. Write this statement as likes(sky, everyone) :- blue(sky). ×
6. ?- X is -(5,3,1). return an error. ✓
7. ?- 3+2 is 5. same as ?- 5 is 3+2. ×

Question.2. MCQ

1. Satellite Image Analysis System is (Choose the one that is not applicable).
   a) Episodic　　b) Semi-Static
   c) Single agent　　d) Partially Observable
2. Categorize Crossword puzzle in Fully Observable / Partially Observable.
   a) Fully Observable　　b) partially Observable
   c) All of the mentioned　　d) None of the mentioned
3. Which were built in such a way that humans had to supply the inputs and interpret the outputs?
   a) Agents　　b) AI system　　c) Sensor　　d) Actuators
4. What is state space?
   a) The whole problem　　b) Your Definition to a problem

1

c) Problem you design d) Representing your problem with variable and parameter

5. What is the main task of a problem-solving agent?

a) Solve the given problem and reach to goal

b) To find out which sequence of action will get it to the goal state

c) All of the mentioned

d) None of the mentioned

6. What is the full form of "AI"?

a) Artificially Intelligent

b) Artificial Intelligence

c) Artificially Intelligence

d) Advanced Intelligence

7. Which of the following is/are Uninformed Search technique/techniques?

a) Breadth First Search (BFS) b) Depth First Search (DFS)

c) Bidirectional Search d) All of the mentioned

8. What is Artificial Intelligence?

a) Artificial Intelligence is a field that aims to make humans more intelligent

b) Artificial Intelligence is a field that aims to improve the security

c) Artificial Intelligence is a field that aims to develop intelligent machines

d) Artificial Intelligence is a field that aims to mine the data

9. Who is the inventor of Artificial Intelligence?

a) Geoffrey Hinton b) Andrew Ng

c) John McCarthy d) Jürgen Schmidhuber

10. Which search implements stack operation for searching the states?

a) Depth-limited search b) Depth-first search

c) Breadth-first search d) None of the mentioned

11. Which of the following is the branch of Artificial Intelligence?

a) Machine Learning b) Cyber forensics

c) Full-Stack Developer d) Network Design

12. What is the goal of Artificial Intelligence?

a) To solve artificial problems

b) To extract scientific causes

c) To explain various sorts of intelligence
d) To solve real-world problems

13. Which of the following is an application of Artificial Intelligence?

a) It helps to exploit vulnerabilities to secure the firm
b) Language understanding and problem-solving (Text analytics and NLP)
c) Easy to create a website
d) It helps to deploy applications on the cloud

14. Breadth-first search always expands the _______ node in the current fringe of the search tree.

a) Shallowest b) Child node
c) Deepest d) Minimum cost

15. Which of the following is a component of Artificial Intelligence?

a) Learning b) Training
c) Designing d) Puzzling

16. Prolog is a programming language based on which paradigm?

a) Imperative programming b) Object-oriented programming
c) Functional programming d) Logic programming

17. In Prolog, knowledge is represented using:

a) Variables b) Loops
c) Facts and rules d) Functions

18. What is the correct syntax for a Prolog rule representing "X is the mother of Y"?

mother(X, Y)

a) mother(X, Y) :- b) X(mother, Y) :-
c) Y is mother(X) :- d) mother(X) = Y :-

19.
```
likes(joe, fish).        likes(joe, mary).
likes(mary, fish).       likes(albert, joe).

?- likes (Who, fish).
```

A) Who = joe.
B) Who = mary.
C) Who = albert.
D) Who = joe. ; Who = mary.

20. What is the function of an Artificial Intelligence "Agent"?
a) Mapping of goal sequence to an action
b) Work without the direct interference of the people
c) Mapping of precept sequence to an action
d) Mapping of environment sequence to an action

21. Depth-first search always expands the _______ node in the current fringe of the search tree.

a) Shallowest
b) Child node
c) Deepest
d) Minimum cost

22. Which of the following machine requires input from the humans but can interpret the outputs themselves?

a) Actuators
b) Sensor
c) Agents
d) AI system

23. Which of the following can improve the performance of an AI agent?

a) Perceiving
b) Learning
c) Observing
d) All of the mentioned

24. Which of the following is/are the composition for AI agents?

a) Program only
b) Architecture only
c) Both Program and Architecture
d) None of the mentioned

25. On which of the following approach A basic line following robot is based?

a) Applied approach
b) Weak approach
c) Strong approach
d) Cognitive approach

26. Which of the following is an example of artificial intelligent agent/agents?

a) Autonomous Spacecraft
b) Human
c) Robot
d) All of the mentioned

27. What is an AI 'agent'?

a) Takes input from the surroundings and uses its intelligence and performs the desired operations
b) An embedded program controlling line following robot
c) Perceives its environment through sensors and acting upon that environment through actuators
d) All of the mentioned

28. Which of the following environment is strategic?

a) Rational
b) Deterministic
c) Partial
d) Stochastic

29. In Prolog, which operator is used to represent logical OR?

a) &&
b) ,
c) !
d) ;

30. In Prolog, which symbol is used to represent the end of a query?

a) !
b) .
c) :
d) ?

31. Which search strategy is also called as blind search?

a) Uninformed search
b) Informed search
c) Simple reflex search
d) All of the mentioned

32. Which search is implemented with an empty first-in-first-out queue?

Fifo
Queue
4
breadth

a) Depth-first search
b) Breadth-first search
c) Bidirectional search
d) None of the mentioned

33. Which search algorithm imposes a fixed depth limit on nodes?

a) Depth-limited search
b) Depth-first search
c) Iterative deepening search
d) Bidirectional search

Abeer Amer

Good Luck

4

Model Examination 1: Foundations and Agent Concepts

True/False Questions (20 points)

1. The Turing Test evaluates whether a machine can exhibit intelligent behavior equivalent to human intelligence. **Answer: True**4
2. In the Chinese Room Argument, understanding is demonstrated through rule-following behavior. **Answer: False** - The argument suggests that rule-following does not constitute understanding4
3. Strong AI hypothesis claims that appropriately programmed computers have cognitive states. **Answer: True**4
4. A fully observable environment provides the agent with access to the complete state of the environment. **Answer: True**1
5. Episodic environments are more complex than sequential environments because agents must consider future consequences. **Answer: False** - Episodic environments are simpler because each episode is independent1
6. The performance measure in PEAS defines what constitutes success for an agent. **Answer: True**1
7. A deterministic environment always produces the same outcome for a given state and action. **Answer: True**1
8. Multi-agent environments only involve competitive relationships between agents. **Answer: False** - They can involve both competitive and cooperative relationships1
9. Goal-based agents consider future consequences of actions when making decisions. **Answer: True**2
10. Reflex agents maintain internal state to handle partially observable environments. **Answer: False** - Simple reflex agents do not maintain internal state2

Multiple Choice Questions (30 points)

11. Which of the following is NOT a component of the PEAS framework?
    a) Performance measure b) Environment c) Actuators d) **Strategy**
    Answer: d) Strategy1
12. An autonomous taxi driver operates in which type of environment?
    a) Single-agent, static b) **Multi-agent, dynamic** c) Single-agent, episodic d) Multi-agent, discrete
    Answer: b) Multi-agent, dynamic1
13. The Total Turing Test includes all of the following capabilities EXCEPT:
    a) Natural language processing b) Computer vision c) **Emotional reasoning** d) Robotics
    Answer: c) Emotional reasoning4

14. In problem formulation, the transition model describes:
    a) The goal state b) **What each action does** c) The initial state d) The path cost
    Answer: b) What each action does2

15. Which environment characteristic describes whether the world changes while the agent deliberates?
    a) Observable b) Deterministic c) **Dynamic** d) Episodic
    Answer: c) Dynamic1

Practical Application Questions (25 points)

16. **Vacuum Cleaner Agent Design (10 points)**
    Design a simple reflex agent for a vacuum cleaner operating in a two-square environment. Specify the agent's percept sequence, action space, and agent function.

Model Answer:

- **Percepts**: [Location (A or B), Status (Clean or Dirty)]
- **Actions**: [Suck, Move Left, Move Right, NoOp]
- **Agent Function**: If current square is dirty → Suck; If current square is clean and location is A → Move Right; If current square is clean and location is B → Move Left12

17. **Environment Classification (15 points)**
    Classify the following environments according to the six environment characteristics:
    a) Chess with a clock
    b) Autonomous driving
    c) Word jumble solver

Model Answer:
a) **Chess with a clock**: Fully observable, strategic, sequential, semi-dynamic, discrete, multi-agent1
b) **Autonomous driving**: Partially observable, stochastic, sequential, dynamic, continuous, multi-agent1
c) **Word jumble solver**: Fully observable, deterministic, episodic, static, discrete, single-agent1

Model Examination 2: Search Strategies and Algorithms

True/False Questions (20 points)

1. Breadth-first search guarantees finding the optimal solution when all step costs are equal. **Answer: True**3
2. Depth-first search has better space complexity than breadth-first search. **Answer: True**3
3. Iterative deepening search combines the benefits of breadth-first and depth-first search. **Answer: True**3
4. Uniform-cost search expands nodes in order of increasing path cost. **Answer: True**5
5. In depth-first search, the frontier is implemented using a queue data structure. **Answer: False** - It uses a stack3

6. Breadth-first search is complete even in infinite state spaces. **Answer: False** - Only complete in finite state spaces3
7. The branching factor affects the time complexity of search algorithms. **Answer: True**3
8. Depth-limited search can distinguish between failure and cutoff. **Answer: True**3
9. Complete search algorithms guarantee finding a solution if one exists. **Answer: True**3
10. Optimal search algorithms always find the least-cost solution. **Answer: True**3

Multiple Choice Questions (30 points)

11. Which search strategy uses LIFO (Last In, First Out) ordering?
    a) Breadth-first b) **Depth-first** c) Uniform-cost d) Best-first
    Answer: b) Depth-first3
12. The time complexity of breadth-first search is:
    a) O(b^d) b) **O(b^(d+1))** c) O(bd) d) O(d^b)
    Answer: b) O(b^(d+1)) where b is branching factor and d is depth3
13. Iterative deepening search performs depth-limited search for:
    a) k = 1, 2, 3 b) **k = 0, 1, 2, ...** c) k = d, d-1, d-2 d) k = random values
    Answer: b) k = 0, 1, 2, ...3
14. Which statement about uniform-cost search is correct?
    a) It's identical to breadth-first search b) **It expands the lowest-cost node first** c) It requires a heuristic function d) It's not optimal
    Answer: b) It expands the lowest-cost node first5
15. The space complexity of depth-first search is:
    a) O(b^d) b) O(bd) c) **O(bm)** d) O(d)
    Answer: c) O(bm) where m is maximum depth3

Practical Application Questions (25 points)

16. **Search Tree Construction (15 points)**
    Given the following graph, show the order of node expansion for both DFS and BFS starting from node A:

text

A connects to B, C

B connects to D, E

C connects to F, G

Model Answer:

- **DFS order**: A, B, D, E, C, F, G (assuming left-to-right tie-breaking)3
- **BFS order**: A, B, C, D, E, F, G3

- **Explanation**: DFS explores depth-first using stack (LIFO), while BFS explores breadth-first using queue (FIFO)3

17. **Algorithm Comparison (10 points)**
Compare DFS and BFS in terms of completeness, optimality, time complexity, and space complexity.

Model Answer:

- **DFS**: Complete (finite spaces), Not optimal, Time O(b^m), Space O(bm)3
- **BFS**: Complete, Optimal (unit costs), Time O(b^(d+1)), Space O(b^(d+1))3
- **Key difference**: BFS guarantees shortest path but requires more memory3

Model Examination 3: Informed Search and Heuristics

True/False Questions (20 points)

1. Heuristic functions must be perfect to be useful in search algorithms. **Answer: False**5
2. Greedy best-first search is guaranteed to find the optimal solution. **Answer: False**5
3. A* search combines path cost and heuristic estimate in its evaluation function. **Answer: True**5
4. The Manhattan distance is an example of an admissible heuristic for the 8-puzzle. **Answer: True**5
5. Informed search strategies use domain knowledge to guide the search process. **Answer: True**5
6. The evaluation function f(n) = g(n) + h(n) is used in A* search. **Answer: True**5
7. Best-first search always expands the node with the lowest f-value. **Answer: True**5
8. Heuristic functions estimate the cost from the current node to the goal. **Answer: True**5
9. Greedy search uses only the heuristic function for node evaluation. **Answer: True**5
10. A* search is both complete and optimal under certain conditions. **Answer: True**5

Multiple Choice Questions (30 points)

11. In A* search, f(n) represents:
a) Path cost b) Heuristic estimate c) **Total estimated cost** d) Branching factor
Answer: c) Total estimated cost5

12. For the 8-puzzle, which heuristic is generally better?
a) Number of misplaced tiles b) **Manhattan distance** c) Euclidean distance d) Random values
Answer: b) Manhattan distance5

13. Greedy best-first search uses which evaluation function?
a) f(n) = g(n) b) **f(n) = h(n)** c) f(n) = g(n) + h(n) d) f(n) = g(n) - h(n)
Answer: b) f(n) = h(n)5

14. An admissible heuristic:
    a) Always overestimates b) **Never overestimates** c) Is always perfect d) Ignores the goal
    Answer: b) Never overestimates5
15. The straight-line distance in map problems is an example of:
    a) An inadmissible heuristic b) **An admissible heuristic** c) A perfect heuristic d) A useless heuristic
    Answer: b) An admissible heuristic5

Practical Application Questions (25 points)

16. **8-Puzzle Heuristic Calculation (10 points)**
    Calculate both heuristics (misplaced tiles and Manhattan distance) for the following 8-puzzle state:

text

Current: [2,8,3]   Goal: [1,2,3]

[1,6,4]   [8, ,4]

[7, ,5]   [7,6,5]

Model Answer:

- **Misplaced tiles**: 6 tiles (2,8,1,6,blank,5 are misplaced)5
- **Manhattan distance**: 2+3+0+1+3+0+3+1 = 135
- **Explanation**: Count horizontal + vertical distances for each misplaced tile5

17. *A Search Trace (15 points)**
    Trace the first three steps of A* search for a simple graph problem, showing f(n), g(n), and h(n) values.

Model Answer:

- **Step 1**: Expand start node, calculate f(n) = g(n) + h(n) for each successor5
- **Step 2**: Select node with lowest f(n) value from frontier5
- **Step 3**: Expand selected node, update frontier with new nodes and their f-values5
- **Key principle**: Always expand the node with minimum f(n) = g(n) + h(n)5

Model Examination 4: Problem Formulation and State Spaces

True/False Questions (20 points)

1. The state space includes all states reachable from the initial state. **Answer: True**2
2. A solution is any path from the initial state to a goal state. **Answer: True**2
3. The transition model describes the result of applying an action in a state. **Answer: True**2

4. Path cost is always equal to the number of steps in the solution. **Answer: False**2
5. The action space defines all possible actions available in every state. **Answer: False** - Actions may vary by state2
6. Goal formulation must precede problem formulation. **Answer: True**2
7. Search trees and state spaces are identical concepts. **Answer: False**2
8. Multiple nodes in the search tree can correspond to the same state. **Answer: True**2
9. The branching factor is determined by the number of possible actions. **Answer: True**2
10. Every search problem has a unique optimal solution. **Answer: False**2

Multiple Choice Questions (30 points)

11. In problem formulation, PEAS stands for:
    a) **Performance, Environment, Actuators, Sensors** b) Problem, Environment, Actions, States c) Path, Evaluation, Actions, Solution d) Planning, Execution, Analysis, Solution
    Answer: a) Performance, Environment, Actuators, Sensors12
12. The 8-queens problem has how many possible board configurations?
    a) 64 b) 8! c) **64!/(64-8)!** d) 2^64
    Answer: c) 64!/(64-8)! (placing 8 queens on 64 squares)2
13. In the Romania map problem, what represents the states?
    a) Actions b) **Cities** c) Roads d) Distances
    Answer: b) Cities2
14. The search tree's root node represents:
    a) The goal state b) **The initial state** c) Any state d) The optimal state
    Answer: b) The initial state2
15. Which is NOT a component of problem formulation?
    a) Initial state b) Actions c) Goal test d) **Search strategy**
    Answer: d) Search strategy2

Practical Application Questions (25 points)

16. **8-Queens Problem Formulation (12 points)**
    Formulate the 8-queens problem by specifying states, initial state, actions, transition model, goal test, and path cost.

Model Answer:

- **States**: Any arrangement of 0 to 8 queens on the board2
- **Initial state**: No queens on the board2
- **Actions**: Add a queen to any empty square2
- **Transition model**: Return the board with the new queen added2

- **Goal test**: 8 queens on board with none attacked2
- **Path cost**: Number of queens placed (minimize conflicts)2

17. **Route Finding Problem (13 points)**
Design a route finding problem for a delivery robot, specifying all problem components and discussing the environment characteristics.

Model Answer:

- **States**: Robot location with package status2
- **Actions**: Move North/South/East/West, Pick up, Drop off2
- **Environment**: Partially observable (limited sensors), stochastic (traffic), dynamic (other vehicles), continuous (real coordinates)12
- **Performance measure**: Minimize delivery time while ensuring safety12
- **Challenges**: Real-time constraints, uncertainty, multiple objectives2

Model Examination 5: Comprehensive Integration

True/False Questions (20 points)

1. Machine Learning is a subset of Artificial Intelligence. **Answer: True**1
2. Data Science and AI are identical fields. **Answer: False**1
3. Reinforcement learning involves rewarding desired behaviors. **Answer: True**4
4. Affective computing deals with emotion recognition and processing. **Answer: True**4
5. The computational complexity of search algorithms depends on the branching factor. **Answer: True**3
6. Bidirectional search can reduce the search complexity significantly. **Answer: True**3
7. Hill climbing is guaranteed to find the global optimum. **Answer: False**3
8. Repeated states in search can lead to infinite loops. **Answer: True**3
9. The choice of heuristic function affects A* search performance. **Answer: True**5
10. All optimal search algorithms have the same time complexity. **Answer: False**3

Multiple Choice Questions (30 points)

11. Which approach to AI focuses on rational decision making?
a) Acting humanly b) Thinking humanly c) Thinking rationally d) **Acting rationally**
Answer: d) Acting rationally4
12. The complexity of breadth-first search in terms of space is:
a) O(b*d) b) **O(b^d)** c) O(d^b) d) O(b+d)
Answer: b) O(b^d)3

13. Which search algorithm is most suitable for problems with large branching factors?
    a) Breadth-first b) **Depth-first** c) Uniform-cost d) Best-first
    Answer: b) Depth-first (better space complexity)3

14. In multi-agent environments, agents can be:
    a) Only competitive b) Only cooperative c) **Both competitive and cooperative** d) Neither competitive nor cooperative
    Answer: c) Both competitive and cooperative1

15. The main advantage of informed search over uninformed search is:
    a) Guaranteed optimality b) **Better efficiency** c) Simpler implementation d) No memory requirements
    Answer: b) Better efficiency5

Practical Application Questions (25 points)

16. **Comprehensive Agent Design (15 points)**
    Design an intelligent agent for an autonomous vacuum cleaner that operates in a house with multiple rooms. Include environment analysis, agent architecture, and performance evaluation.

Model Answer:

- **Environment Analysis**: Multi-room (partially observable), stochastic (debris distribution), sequential (cleaning sequence matters), dynamic (people moving), continuous (navigation), single-agent1
- **Agent Architecture**: Goal-based agent with mapping capability, path planning, and battery management12
- **Performance Measure**: Percentage of area cleaned, energy efficiency, time to completion1
- **Sensors**: Cameras, proximity sensors, dirt detectors1
- **Actuators**: Wheels, vacuum motor, brush system1

17. **Search Algorithm Selection (10 points)**
    Given a search problem with the following characteristics: large state space, high branching factor, solutions exist at various depths, and memory is limited. Recommend the most appropriate search algorithm and justify your choice.

Model Answer:

- **Recommendation**: Iterative Deepening Depth-First Search (IDDFS)3
- **Justification**:
    - Low memory requirements like DFS (O(bm))3
    - Complete like BFS3
    - Optimal for unit step costs3

- Handles large branching factors better than BFS3

- **Alternative**: If heuristic available, use IDA* (Iterative Deepening A*)

Model Examination 6: Adversarial Search and Game Theory

True/False Questions (20 points)

1. In adversarial search, the minimax algorithm assumes both players play optimally. **Answer: True**
2. Alpha-beta pruning can reduce the number of nodes evaluated without affecting the final decision. **Answer: True**
3. The evaluation function in game trees always returns exact utility values. **Answer: False** - It estimates positional advantage.
4. Monte Carlo Tree Search (MCTS) relies solely on heuristic evaluation. **Answer: False** - Uses random simulations.
5. The "horizon effect" occurs when a player delays inevitable outcomes. **Answer: True**
6. Iterative deepening is only useful in real-time strategy games. **Answer: False** - Applicable to any game with time constraints.
7. The Nash equilibrium concept applies only to zero-sum games. **Answer: False** - Relevant to all non-cooperative games.
8. Transposition tables store previously computed game states to avoid redundancy. **Answer: True**
9. Expectiminimax extends minimax to games with chance elements. **Answer: True**
10. In partially observable games, belief states represent probabilistic knowledge. **Answer: True**

Multiple Choice Questions (30 points)

11. Which algorithm uses two values to prune game trees?
    a) Minimax
    b) **Alpha-beta pruning**
    c) A*
    d) BFS
    Answer: b) Alpha-beta pruning
12. The minimax algorithm's time complexity is:
    a) O(b^d)
    b) **O(b^(d/2))**
    c) O(d^b)
    d) O(bd)
    Answer: a) O(b^d)
13. In a game tree, a "quiescent" state is one where:
    a) **No drastic value changes are expected**
    b) The game is tied

c) Both players cooperate
d) All moves lead to terminal states
Answer: a) No drastic value changes

14. Which technique addresses the "horizon problem"?
a) Iterative deepening
b) **Forward pruning**
c) Heuristic evaluation
d) Transposition tables
Answer: b) Forward pruning

15. The primary purpose of an evaluation function is to:
a) **Estimate the utility of non-terminal states**
b) Determine optimal moves
c) Calculate exact game outcomes
d) Initialize game trees
Answer: a) Estimate utility

Practical Application Questions (25 points)

16. **Minimax Algorithm Trace (15 points)**
Apply minimax to the following game tree. Indicate propagated values and the optimal path.

text

```
    A
   / | \
  B  C  D
 / \    |
E  F    G
3  5    2
```

Model Answer:

- B: min(E=3, F=5) → 3
- D: min(G=2) → 2
- A: max(B=3, C=?, D=2) → 3 (assuming C's children not shown)

17. **Alpha-Beta Pruning (10 points)**
Identify pruned nodes in a game tree when using alpha-beta with node order: A → B → D → C.
Explanation: Nodes under C are pruned if B's value exceeds C's current alpha.

Model Examination 7: Clustering with k-Means

True/False Questions (20 points)

1. k-means guarantees convergence to the global optimum. **Answer: False** - Local optima possible.
2. The elbow method determines optimal k by minimizing inertia. **Answer: True**
3. Feature scaling is unnecessary for k-means. **Answer: False** - Required for distance metrics.
4. k-means++ initializes centroids to improve convergence. **Answer: True**
5. Silhouette scores range from -1 to 1. **Answer: True**

Multiple Choice Questions (30 points)

11. The k-means objective function minimizes:
    a) **Sum of squared distances**
    b) Manhattan distance
    c) Entropy
    d) Covariance
    Answer: a) Sum of squared distances

12. Which metric evaluates cluster separation?
    a) **Silhouette coefficient**
    b) ROC-AUC
    c) F1-score
    d) Precision
    Answer: a) Silhouette coefficient

13. For high-dimensional data, k-means may suffer from:
    a) **Curse of dimensionality**
    b) Overfitting
    c) Underfitting
    d) Multicollinearity
    Answer: a) Curse of dimensionality

Practical Application Questions (25 points)

16. **k-Means Iteration (15 points)**
    Given points (2,3), (4,5), (6,7), and initial centroids (3,4) and (5,6), compute one iteration.
    Model Answer:

- Cluster 1: (2,3), (4,5) → New centroid (3,4)
- Cluster 2: (6,7) → New centroid (6,7)

Model Examination 8: Classification with Naive Bayes

True/False Questions (20 points)

1. Naive Bayes assumes feature independence. **Answer: True**

2. Laplace smoothing handles zero-probability cases. **Answer: True**
3. Naive Bayes is unsuitable for text classification. **Answer: False** - Widely used in NLP.

Multiple Choice Questions (30 points)

11. The Naive Bayes classifier applies:
    a) **Bayes theorem**
    b) Central limit theorem
    c) Law of large numbers
    d) Markov assumption
    Answer: a) Bayes theorem

12. For continuous features, which distribution is commonly assumed?
    a) **Gaussian**
    b) Poisson
    c) Binomial
    d) Exponential
    Answer: a) Gaussian

Practical Application Questions (25 points)

16. **Naive Bayes Calculation (15 points)**
    Predict class (Spam/Not Spam) for "Win money now" with prior P(Spam)=0.3 and likelihoods P("win"|Spam)=0.1, P("money"|Spam)=0.2, P("now"|Spam)=0.05.
    Model Answer:

- P(Spam|text) ∝ 0.3 * 0.1 * 0.2 * 0.05 = 0.0003
- Assume Not Spam probability higher → Classify as Not Spam.

Model Examination 9: Probabilistic Reasoning

True/False Questions (20 points)

1. Bayesian networks encode conditional dependencies. **Answer: True**
2. Markov assumption implies dependency on all prior states. **Answer: False** - Only immediate past.

Multiple Choice Questions (30 points)

11. Hidden Markov Models (HMMs) are used for:
    a) **Temporal pattern recognition**
    b) Clustering
    c) Dimensionality reduction
    d) Feature selection
    Answer: a) Temporal pattern recognition